

Highlights

Evaluation geometry and the limits of cross-region transfer in pre-fire thermal wildfire prediction

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- Scoring the same model at the fire, not region-wide, costs 0.143 AUC
- The cause is the negative pool, not prevalence: +0.155 against -0.000
- Thermal predictors add +0.06 to +0.15 AUC within all five fire regions
- Equalising the frame lifts transfer 0.540 to 0.617, six below chance to one
- Of 20 pre-deployment diagnostics, none that is label-free orders transfer

Evaluation geometry and the limits of cross-region transfer in pre-fire thermal wildfire prediction

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Abstract

Pre-fire thermal dryness separates a fire year from a normal year, but models built on it are rarely tested outside the region where they were fitted. Five Mediterranean wildfire regions were analysed on about 500 m cells, with MCD64A1 labels and spatially blocked validation, adding six thermal predictors to a terrain, fuel and greenness baseline.

Local skill does not travel, and most of it is lost before the region changes. Within-region ROC-AUC rose by +0.056 to +0.153 under blocked cross-validation, but by +0.022 [−0.032, +0.077] when a whole burn scar is withheld, and by +0.004 [−0.028, +0.036] across twenty transfer directions, with a sign that changes from pair to pair. The static baseline transferred no better, 0.537 against 0.541, so this is not a property of the dynamic block. On **identical cells**, a model holding the held-out scar scores 0.634, one without it 0.552 and one fitted 306 to 2,802 km away 0.555, so neither withholding the fire nor crossing the region is shown to cost anything. What is measurable is the evaluation area itself, worth **0.143** [+0.077, +0.208] of the fall from a region-wide 0.776, because its negatives are all fire-adjacent and therefore

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the hardest in the region.

The failure is invisible before deployment. Of twenty candidate diagnostics only two had intervals excluding zero; both measure sign agreement over a selected feature subset ($\rho = 0.84$, post-selection) and both need target labels. No marginal measure ordered the matrix, so the family that works is the one a practitioner does not have.

Most of the apparent mechanism is an artefact of the evaluation frame. Five predictors reverse direction between regions on the frames as drawn, but those frames enclose very different far fields, from 2 % to 63 % of cells beyond 10 km of any fire, on higher ground. A 10 km collar, which drops no burned cells, makes all five agree in sign on elevation, LST and TVDI, lifts transfer from 0.540 to 0.617, drops the directions below chance from six to one, and leaves no reversal bootstrap-supported under our own criterion. The agreed sign is not the one the dryness framing predicts: hotter surfaces burned less everywhere, surviving stratification within elevation and greenness.

Label-free alignment pushed fourteen of twenty directions towards chance rather than repairing them. Transfer skill has to be measured, not inferred from similarity.

Keywords: wildfire, model transferability, area of applicability, land surface temperature, domain adaptation, concept shift

1. Introduction

Wildfire is a defining disturbance of Mediterranean-basin landscapes, and a changing climate is reshaping where and how it burns [29]. The pressure

to anticipate *where* fire will occur has produced a large and still-growing literature on fire susceptibility mapping. It has recently been synthesised from both a methodological and an application perspective [42, 14]. The dominant pattern is well established. Geospatial predictors are assembled over a study region and paired with a historical record of burned area. A supervised classifier is then fitted, most often a random forest [5]. The resulting susceptibility surface is published with a cross-validated AUC in the 0.85 to 0.95 range.

This paper is about a property that the pattern does not report. Predictors that describe the *state* of a surface in a particular season are more informative about that season than predictors that describe a place. They are also, as we show, not portable: the relationship they carry is reparameterised locally, existing everywhere but pointing in different directions in different places. The dynamic block is worth +0.056 to +0.153 ROC-AUC inside every region under blocked cross-validation, +0.022 when a whole burn scar is withheld, and contributes +0.004 between them, an estimate whose interval spans zero, with a sign that is a property of the source-target pair rather than of the block.

The obvious reading of that pair of numbers is a trade-off, and our own earlier framing took it. The data do not support it. The static baseline transfers at a mean of 0.537 against the thermal model’s 0.541, so the predictor class that ought to travel does not travel either, and removing the reversing predictors returns +0.014 of transfer on an interval that also spans zero. Two nulls on the portability axis are not an exchange. What is measured is a local gain that does not export, and a local cost when the reversing predictors are

removed, with no compensating transfer gain on either arm.

30 *1.1. Portability goes unmeasured*

A susceptibility model’s reported skill is almost always an estimate of *within-region* performance. Held-out folds come from the same study area and season, often from the same fire event, and where folds are drawn at random over cells, spatial autocorrelation inflates the estimate further. The
35 problem is documented across ecological modelling [33, 31] and is addressed by spatially blocked cross-validation [39, 25]. Blocking removes the inflation that random folds produce, and it is the right correction for what it corrects. It does not make the estimate honest about a fire the model has not seen. Section 4.3 measures that on identical cells: a model with the held-out burn
40 scar in its training data scores 0.634 on that scar’s area, and withholding the scar gives 0.552. Part of the apparent gap between a region-level figure and either of those is the evaluation area rather than the model.

Because only that side of the ledger is reported, portability is never entered at all. A predictor block is adopted on the strength of the increment
45 it delivers inside its training footprint, and whether that increment survives a change of region is not asked, even though any regional product built from locally trained models implicitly promises generalisation beyond that footprint. For fire specifically the transfer question is asked far less often than in adjacent fields. Meteorological fire-danger indices are known not
50 to port cleanly between fire environments [32], and the two recent studies that test model transfer systematically, across Mediterranean countries [8] and US counties [16], both report that it largely succeeds between similar regions. Both transfer models whose dominant predictors are *spatially sta-*

tionary: terrain, human modification, night-time lights, population density,
55 long-term climatologies. Each describes a place rather than a season, which is
the class our thesis expects to travel well. Species distribution modelling has
examined transferability far more systematically [46], and there geographic
and environmental similarity do *not* reliably predict transfer success [41, 34].

1.2. *Pre-fire thermal dryness is the natural test case*

60 The dynamic class used here is pre-fire thermal dryness, and it is the
class most plausibly *expected* to transfer. Standard susceptibility predictors
are terrain, fuel type, greenness, long-term climatology and proximity to
roads and settlements, and most are static or near-static over the timescale
at which fire danger varies. Such a set explains poorly why a particular sum-
65 mer burned and the preceding one did not. What changes between those
summers is the state of the surface: fuel dryness, moisture stress, and how
anomalously hot the land surface has become relative to its own baseline.
Satellite thermal observation gives direct access to part of that state through
fuel moisture content [47], and pairing surface temperature with a vegeta-
70 tion index was established as a live-fuel-moisture estimator for fire-danger
rating by Chuvieco et al. [7]. The Temperature-Vegetation Dryness Index
[35] formalises that feature space into an internally normalised measure. In
principle it should be less exposed to absolute-temperature offsets between
regions than raw land surface temperature. That theoretical portability ad-
75 vantage is tested here, and it is not found: the two internally normalised
channels transfer no better than the four absolute ones, at means of 0.544
and 0.548 over the twenty directions (Appendix A(f)).

That pre-fire thermal state carries genuine information about subsequent

fire is established [19, 21, 20, 9]. What has not been established is whether
80 the predictive *skill* of a classifier trained on it in one fire region survives
strict, label-free application to another. The physics linking moisture stress
to combustion is universal, so portability should be most expected for this
class, which is what makes it diagnostic: a loss here cannot be dismissed as a
peculiarity of a locally defined covariate. That expectation is the motivation
85 for the design, not a finding of it. Section 4.9 reports that the signed associ-
ations run the other way, a hotter pre-fire surface being associated with less
burning in all five regions and surviving stratification within elevation and
greenness, so on this cohort the block behaves as a proxy for fuel availability
rather than as a moisture-stress index. We keep the framing because it is
90 why these predictors were chosen and why their failure to travel is informa-
tive, and we state the contradiction where the evidence appears rather than
adjusting the motivation after the fact.

1.3. Why the loss is invisible to the diagnostics in use

A deficit on transfer can arise two ways. Under **covariate shift** the
95 marginal distribution of the predictors differs while the predictor-response
relationship is preserved, which is in principle correctable without target la-
bels, by per-region standardisation or covariance alignment such as CORAL
[37], an approach with an established remote-sensing literature [38]. Under
concept shift the conditional relationship itself changes: a predictor pos-
100 itively associated with burning in one region is negatively associated with
it in another. No label-free realignment of input distributions can repair
a sign reversal, because the information needed to detect it is exactly the
information being withheld. The taxonomy is canonical [27].

This distinction is what makes the failure hard to anticipate. The di-
 105 agnostics the field relies on to decide whether a model may be applied to
 new ground, above all area-of-applicability measures built on dissimilarity in
 predictor space [23, 24, 18], are marginal by construction. A target region
 can sit well inside the training data’s predictor envelope while the relation-
 ship between those predictors and burning points the other way. This paper
 110 measures both sides on the same pairs.

1.4. Contributions

Three findings carry this paper. Each is stated at the strength its interval
 supports, and the nearest prior work is named inside the claim.

Contribution 1. Where a model is scored decides what it ap-
 115 **pears to know, and the effect is large enough to dissolve one of our**
own findings. That evaluation extent inflates AUC is established in species
 distribution modelling, by Lobo et al. [17], VanDerWal et al. [40] and Barve
 et al. [3] (Section 2.5); what is new here is a magnitude, under a controlled
 design that earlier work did not run, and its consequences for a live wildfire
 120 result. Holding the model, the predictors and the fitting fixed and changing
 only which cells are scored, moving from the whole region to the burn scar
 and its 2 km collar costs **0.143 ROC-AUC** [**+0.077, +0.208**], from 0.776
 to 0.634. That exceeds the predictor-block increments this literature pub-
 lishes as findings, so it is not a second-order correction. A control isolates
 125 the cause: matching a region sample to the scar area’s prevalence changes
 nothing, at -0.000 [$-0.003, +0.002$], while scoring the same predictions on
 the scar area costs $+0.155$ [$+0.093, +0.217$]. It is the composition of the
 negative pool, whose cells are all fire-adjacent and therefore the hardest in

the region, not class balance.

130 Applied between regions, the same effect withdraws three of our own
claims (Section 4.9). The five study areas enclose very different far fields,
from 2 % of Montiferru’s cells lying beyond 10 km of any fire to 63 % of Bejís’s,
and that far field is higher ground. Restricting every region to a 10 km collar,
which drops no burned cells, makes all five agree in sign on elevation, on
135 LST and on TVDI; both bootstrap-supported elevation reversals disappear;
mean transfer rises from 0.540 to 0.617; and the directions below chance
fall from six to one. The sign reversal we had offered as the mechanism
of the residual is therefore mostly a property of how five rectangles were
drawn. Under this paper’s own criterion **no reversal remains bootstrap-**
140 **supported** in the equalised frame; the closest is the LST anomaly between
Bejís and Evia, a point reversal whose Evia interval includes 0.5 by 0.003.
The sign the five regions now agree on is also not the one the dryness framing
predicts: hotter pre-fire surfaces burned less in every region, and that survives
stratification within elevation and greenness deciles, so on this cohort the
145 block behaves as a proxy for fuel availability rather than as a moisture-
stress index (Section 4.9). A reversal also appears between two fires eleven
months apart inside one study area, where the frame is fixed by construction
(Section 4.8), but on elevation alone, on eleven positive-carrying blocks, and
with one fire per arm it is as consistent with event-to-event variation as with
150 any regional property. The practical consequence is a reporting standard,
stated in Section 5.8.

**Contribution 2. Local skill does not travel, and correcting the
frame does not rescue it.** The thermal block is worth +0.056 to +0.153

ROC-AUC inside every one of five regions under blocked cross-validation,
 155 with every bootstrap interval above zero and the result holding in both analysis populations. Much of that is a property of interleaved holdout: withholding a whole burn scar leaves $+0.022$ $[-0.032, +0.077]$. Across twenty ordered transfer directions it contributes $+0.004$ $[-0.028, +0.036]$, an interval spanning zero under all four between-direction resampling units we
 160 computed, with a sign that varies by pair — paired deltas run from -0.148 to $+0.133$, twelve positive and eight negative, and dropping Evia alone reverses their mean. Two controls bound the reading. The static baseline transfers at 0.537 against the thermal model’s 0.541 , so the portability failure is not the dynamic block’s peculiarity. And equalised transfer of 0.617
 165 against within-region skill of about 0.87 leaves most of the gap intact, so the frame correction of Contribution 1 narrows the mechanism without rescuing the result. The within-region half is not novel, comparable results existing for these landscapes [1, 13]; the paired contrast against portability is. Dimarco et al. [8] transfer a predominantly stationary predictor set successfully
 170 across a comparable Mediterranean design, though their response variable is ignition rather than burned area (Section 5.7).

Contribution 3. The shortfall is invisible to every diagnostic that could be run before deployment. Twenty candidate diagnostics from five families are evaluated against observed transfer under one
 175 bootstrap framework. Area-of-applicability-style predictor-space dissimilarity [23, 24, 18], climatic and geographic distance, learned domain separability, niche-overlap statistics and burn-pattern regime distances were none of them shown to order the matrix; the domain classifier is at ceiling, with $AUC \geq$

0.96 for every pair. Only two diagnostics have intervals excluding zero, and
 180 both measure conditional direction agreement, that is, whether each pre-
 predictor’s signed association points the same way in both regions (Spearman
 $\rho = +0.84$ and $+0.81$ over sixteen directions from eight pairs). **Both are
 computed on a data-selected subset of predictors, and their all-
 nine-feature counterparts span zero, so the result rests on that**
 185 **selection.** Signed associations need burned labels on both sides, so this is
 a mechanism diagnosis rather than a screen: the marginal family, the only
 one runnable before deployment, is the one that fails. The same point sur-
 vives without any ranking, at the point estimates: the pair with the highest
 burned-niche overlap fails in both directions while the lowest transfers in
 190 both. That contradicts an expectation the fire literature carries [8, 16] and
 agrees with species distribution modelling [41, 34].

Two consequences follow, reported as supporting results rather than lead-
 ing ones. Label-free alignment by standardisation and covariance alignment
 [37] does not repair transfer; it compresses fourteen of twenty directions
 195 towards chance, and we believe this is the first application of covariance
 alignment to fire susceptibility. And because the residual is conditional, the
 resource that closes it is target labels: thirty-two labelled 5 km blocks recover
 85 to 89 % of the target’s matched ceiling in three of six directions, 51 to 57
 % in two more, and 30 % in the sixth.

200 Alongside these, a leakage-audited, spatially blocked evaluation and trans-
 fer protocol is released with code, configuration and frozen outputs, so that
 most of this result can be re-run rather than taken on trust. One component
 is an exception and is named in the data-and-code availability statement:

the reproduction-check code is not in the released repository at the commit
205 of record. The few-shot export, previously also listed as an exception, was
regenerated at commit 6d7a6a71 and its validator now passes on every check.
This matters given evidence that wildfire transfer conclusions are sensitive
to evaluation design and task formulation [45].

A companion paper treats the observational layer beneath this one, where
210 preprocessing decisions taken before any model is fitted are shown to carry
their own uncertainty budget.

Section 2 reviews the relevant literature. Section 3 describes the study
regions, the data, and the modelling, cross-validation, transfer and adapta-
tion protocol. Section 4 reports results. Section 5 discusses them, including
215 limitations. Section 6 concludes.

2. Related work

2.1. Fire susceptibility mapping, and what it reports

Fire susceptibility mapping has been synthesised recently from both method-
ological and application perspectives [42, 14]. The dominant pattern is sta-
220 ble: geospatial predictors are assembled over a study region, paired with a
historical burned-area record, and fitted with a supervised classifier, most
often a random forest [5], and the resulting surface is published with a cross-
validated AUC in the 0.85 to 0.95 range. That figure is a within-region
estimate. Comparable within-region results already exist for the landscapes
225 studied here [1, 13].

2.2. *Pre-fire thermal dryness*

Satellite thermal observation gives repeated access to surface state through fuel moisture content [47]. Pairing land surface temperature with a vegetation index was established as a live-fuel-moisture estimator for fire-danger rating by Chuvieco et al. [7], and the Temperature-Vegetation Dryness Index [35] formalises that feature space into an internally normalised measure, which is why it is often expected to travel between regions better than raw temperature. That pre-fire thermal state carries information about subsequent fire is established [19, 21, 20]. Gelabert et al. [9] found dead fine fuel moisture and its anomalies to be the most influential predictor of human-caused ignition likelihood across Europe, testing generalisation by pooled fitting with per-site evaluation. What has not been tested is whether a classifier built on this class survives strict, label-free application to a region it never saw.

2.3. *Spatial validation and transferability*

Random cross-validation over spatially autocorrelated cells inflates skill estimates [33, 31], and spatially blocked designs are the standard remedy [39, 25]. The remedy is not uncontested: blocked schemes have been argued to introduce pessimistic bias of their own [43, 26, 6]. This paper uses spatial blocking throughout and reports every result at three block sizes, so that a reader can see which verdicts depend on the choice.

Blocking corrects the inflation that random folds produce within a region. It does not speak to performance on a fire the model has not seen, which Section 4.3 measures on identical cells. For that, the field’s instrument is the area of applicability and related predictor-space dissimilarity measures

[23, 24, 18], which ask whether a target’s predictor values fall inside the training data’s envelope. Species distribution modelling has examined transferability far more systematically than fire has [46], and two studies there report that geographic and environmental similarity do not reliably predict transfer success [41, 34], and the transferability of satellite-derived empirical models is an active question in fire remote sensing itself [22].

2.4. Dataset shift and label-free adaptation

The canonical taxonomy separates covariate shift, where the predictor distribution moves but the predictor-response relationship holds, from concept shift, where the relationship itself changes [27]. Covariate shift is in principle correctable without target labels, by per-region standardisation or by covariance alignment such as CORAL [37], and domain adaptation has an established remote-sensing literature [38, 30]. Concept shift is not: no realignment of input distributions can repair a reversal in the sign of an association, because detecting it requires the labels being withheld. Shift decomposition in applied remote sensing is not itself new [12]. We are not aware of a prior application of covariance alignment to fire susceptibility, fire occurrence or burned-area prediction.

2.5. Cross-region generalisation of fire models

Few studies test fire-model transfer directly. Podschwit et al. [32] report, in a Peruvian case study, that meteorologically derived danger indices do not port cleanly between fire environments. WildfireGenome [16] runs a leave-one-county-out matrix across seven US counties and reports strong within-county performance with highly variable off-diagonal transfer, similar

275 pairs transferring well and dissimilar pairs collapsing; its label, however, is
a principal-component composite of hazard *indicators* rather than observed
burned area, and it applies no adaptation. Xu et al. [45] argue that wild-
fire transfer conclusions depend strongly on evaluation design, a caution we
address by fixing the protocol in a project log before the diagnostics were
280 computed and by reporting every sensitivity axis; that log is not a formal pre-
registration, and Section 3.11 states what was fixed and when. Kondylatos
et al. [15] provide Mesogeos, a 1 km Mediterranean datacube for data-driven
wildfire modelling.

Evaluation extent and AUC. Species distribution modelling settled
285 long ago that the area a model is evaluated over is not a neutral choice. Lobo
et al. [17] make it the fifth and, by their own ranking, most important rea-
son to distrust AUC as a comparative measure: the extent of the modelled
area governs how many easy absences enter the calculation, and therefore
the score. VanDerWal et al. [40] show the same lever on the calibration side,
290 that how far background points are drawn from known presences changes
the fitted model and its apparent accuracy, and Barve et al. [3] give the ar-
gument its general form as the accessible area. This literature is qualitative
about magnitude, because magnitude is problem-specific: it establishes that
the effect exists, is directional, and can be large, without fixing a number for
295 any given application. Two things follow for the present paper. First, our
Contribution 1 is a measurement inside that framework rather than a new
phenomenon, and Section 1 states it that way. Second, the wildfire literature
has largely not imported the lesson: region-wide validation figures are re-
ported as though they described performance at the fire, and we know of no

300 wildfire study that holds the model fixed and varies only the evaluation cells.
The number we obtain, 0.143 ROC-AUC, is larger than the predictor-block
increments this literature reports, which is why we treat it as a reporting
problem in Section 5.8 rather than a caveat.

The nearest neighbour, and the contrast the present paper draws.

305 Dimarco et al. [8] is the closest Mediterranean analogue: 500 m predic-
tors harmonised across four countries, tree ensembles under spatial cross-
validation, transfer tested both leave-one-country-out and as a full 4 x 4
matrix, with every transfer exceeding AUC 0.80 and bioclimatically similar
countries scoring higher. They apply no domain adaptation.¹

310 Much of the design is shared — Mediterranean regions, 500 m cells,
MCD64A1-derived targets, tree ensembles, spatially aware validation, an ex-
plicit transfer matrix — and two things differ. **The predictor class:** their
model rests on attributes of a place, all spatially stationary, ours on the state
of a surface in one season. **The response variable:** theirs is human-driven
315 ignition, ours is burned area, and those are different quantities with different
dominant controls. Read together the two results suggest that the relation-
ship between domain similarity and transfer success may be predictor-class
dependent, with Vesk et al. [41] and Rousseau and Betts [34] aligning from
species distribution modelling while Dimarco et al. and WildfireGenome run
320 against it. We present that as a live disagreement, and Section 5.7 states
why our own data do not settle it.

Methods, where no land surface temperature or TVDI variable appears

¹Their Results text refers to "LST anomalies", which is inconsistent with their own

and the only temperature predictor is a static ERA5-Land seasonal climatology [28]. We follow their Methods, which we take to be authoritative, and
 325 note the discrepancy so that the comparison is transparent. Every statement about their work was checked against the full text, read 2026-08-08.

3. Data and methods

3.1. Study regions and temporal windows

Five Mediterranean wildfire regions are analysed (Fig. C.1): Manavgat
 330 2021 and Muğla 2021 in Türkiye, Bejís 2022 in Spain, North Evia 2021 in Greece and Montiferru 2021 in Sardinia. Each is a place-based rectangular area of interest in EPSG:4326, defined from place coverage rather than from a fire perimeter, and not tuned on burned prevalence, gate outcome or any model metric. One AOI choice was label-informed and is stated as such.
 335 The North Evia box was extended after the legacy box was found to carry an atypically high burned prevalence. The extended geometry was then defined from place anchors, and the legacy variant is kept as a sensitivity arm (Appendix A). The AOIs are deliberately not clipped to fire perimeters, so that unburned cells around each fire form the negative class rather than
 340 being excluded by construction. A sixth region, Kozan 2023, is carried as a negative control and is excluded from all modelling by the gate of Section 3.3.

Each region has two non-overlapping windows. The **predictor window** closes the day before the **label window** opens, so no predictor observation can post-date the first labelled burning. Window lengths vary between re-
 345 gions with the event: 57 to 61 days for predictors and 35 to 59 days for labels, counted inclusively. A four-year baseline of window-symmetric composites

precedes each predictor window and supplies the climatological reference for the anomaly channels. Region identifiers, bounding boxes, window dates and baseline years are registered in `core/regions.py` and are reproduced here.

Table 1: Study regions, areas of interest and temporal windows. Bounding boxes are in EPSG:4326, as registered. Window lengths in brackets are inclusive day counts. The baseline years are the four window-symmetric years preceding each predictor window.

Region	Bounding box (lon min, lat min, lon max, lat max)	Predictor window	Label window	Baseline years
Manavgat 2021 (Türkiye)	31.05, 36.72, 31.85, 37.35	2021-06-01 to 2021-07-27 (57 d)	2021-07-28 to 2021-08-31 (35 d)	2017, 2018, 2019, 2020
Bejís 2022 (Spain)	-1.05, 39.68, -0.35, 40.15	2022-06-15 to 2022-08-14 (61 d)	2022-08-15 to 2022-09-30 (47 d)	2018, 2019, 2020, 2021
Muğla 2021 (Türkiye)	27.1, 36.6, 28.9, 37.45	2021-06-01 to 2021-07-28 (58 d)	2021-07-29 to 2021-09-15 (49 d)	2017, 2018, 2019, 2020
North Evia 2021 (Greece)	23.05, 38.55, 23.85, 39.15	2021-06-05 to 2021-08-02 (59 d)	2021-08-03 to 2021-09-30 (59 d)	2017, 2018, 2019, 2020
Montiferru 2021 (Italy)	8.45, 40.05, 8.75, 40.27	2021-05-25 to 2021-07-23 (60 d)	2021-07-24 to 2021-08-31 (39 d)	2017, 2018, 2019, 2020

3.2. Burned-area label and the ~ 500 m analysis grid

Labels come from MODIS MCD64A1 Collection 6.1 [10] retrieved through Google Earth Engine [11], whose omission and commission characteristics [4] bound every model here. The analysis grid is **reconstructed rather than native**: the pipeline’s 30 m EPSG:4326 reference grid is partitioned into 17 x 17 blocks, giving a nominal 510 m cell that approximates rather than reproduces the MODIS cell, and is square in degrees but not on the ground. A cell’s burn date is the mode of its positive sub-pixel day-of-year values, tested against the region’s label window; the label never affects a cell’s eligibility for

modelling. Two safeguards are recorded rather than assumed: cells burning
360 inside a region’s own predictor window are removed, which ran for three of
five regions, and burning in earlier years is screened for none. Appendix C.1
gives the full specification, including what follows from the grid’s shape.

3.3. *Burned-landcover admissibility gate*

Before any modelling each region passes a gate asking what fraction of
365 its burned cells is dominated by natural vegetation, using ESA WorldCover
classes on the same cells. A region is admitted at 0.50 with at least 30
burned cells, and rejected as a cropland-dominated control at 0.50 cropland.
The purpose is to separate burned area produced by natural-fuel combustion
from post-harvest stubble burning, which MCD64A1 does not distinguish.
370 Verdicts are in Section 4.1; the full rule is in Appendix C.1.

3.4. *Predictor variables*

Ten predictors are used, four baseline and six thermal, all summarised
per cell over the predictor window, through the chain shown in Fig. C.2.
All optical and thermal predictors come from Landsat 8 Collection 2 Level-2
375 (LANDSAT/LC08/C02/T1_L2), quality-screened per pixel from the QA_PIXEL
band with the water bit deliberately preserved; the coarse-resolution thermal
input is MODIS/O61/MOD11A1.

- **NDVI**, the predictor-window median of Landsat surface reflectance, is
the baseline’s one time-varying member.
 - **Elevation** and **slope** come from the Copernicus DEM GLO-30.
- 380

- **Land cover** is ESA WorldCover v200 [48], entering as the dominant class code.
- **Current LST**, the predictor-window median of Landsat surface temperature in °C.
- 385 • **LST anomaly**, a z-score of the current-window LST median against the four baseline years, set to no-data where the baseline standard deviation is below 1.0 °C or observations are too few.
- **TVDI and TVDI difference**. The Temperature-Vegetation Dryness Index [35] is computed in the LST-NDVI feature space, with wet and
390 dry edges taken as the 2nd and 98th LST percentiles within each of twenty NDVI bins and the index clamped to [0, 1]. The edges are percentiles of the LST values a given scene contains, so they are fitted per AOI and per window, and a TVDI of 0.5 denotes a different physical moisture state in each region. The companion paper reports
395 what that scene dependence does and does not explain. **TVDI difference** is the raw anomaly of that index against the same four window-symmetric baseline years used for the LST anomaly: `tvdi_difference = current_tvdi - mean(baseline_tvdi)`, in index units rather than standard deviations, which is why it is reported alongside the z-scored
400 channel rather than in place of it.
- **Downscaled LST and fused LST**. A MODIS-to-Landsat downscaling model is trained on the predictor window and applied to the full 30 m grid. The fused product equals observed Landsat LST wherever that is valid, and the downscaled surface only where it is not. Gap-filling

405 therefore never replaces or blends a valid observation. The gap-filled
share is 0.11 % to 9.70 % by region. The downscaler’s own inputs
include coordinates, which is the one route by which a coordinate-
derived surface re-enters a feature set from which Section 3.14 excludes
coordinates. Appendix A(g) reports the increment without these two
410 channels.

3.5. Cell aggregation, validity and analysis populations

Cell-level values are means over the ~ 510 m cell, computed from valid
30 m pixels only, with the valid fraction recorded. A cell is `valid_for_`
`modeling` when it meets a 30 % valid-pixel floor and no thermal channel
415 is missing. Two populations are then defined. The **primary** population
is natural vegetation, cells whose combined tree, shrub and grass fraction
reaches 0.50, which excludes cropland from every burnable mask. The **sec-**
ondary population is all valid cells, reported throughout as a sensitivity.
Every headline result is given for both.

420 3.6. Feature sets and classifier

Two nested feature sets are compared. The **baseline** set is NDVI, el-
evation, slope and land cover. The **thermal** set is the baseline plus the
six thermal channels. Land cover is one-hot encoded and numeric features
are standardised inside the fitting pipeline. The classifier is a random for-
425 est [5] with 300 trees, unlimited depth, `min_samples_leaf = 3`, balanced
class weights and `random_state = 42`, identical for every region, popula-
tion, feature set and transfer direction, so that no comparison in this paper
is confounded by a model choice.

3.7. *Spatial-block cross-validation and bootstrap uncertainty*

430 Cross-validation is spatially blocked. Blocks are squares of the analysis grid formed as `row_500m // B` by `col_500m // B`, and folds are drawn over whole blocks, so no block is split between training and test. Five folds are used. The block size B is reported at 2, 10 and 20 cells, about 1, 5 and 10 km, for the within-region analysis.

435 Uncertainty is a spatial-block bootstrap: blocks are resampled with replacement, 1000 replicates, seed 42, with 2.5 and 97.5 percentiles as the interval. Because it is blocks that are resampled, what bounds an interval’s reliability is the number of blocks carrying at least one burned cell, and those counts are reported alongside the intervals. Where a verdict rests on too few
440 such blocks it is stated as indicative rather than as an interval.

3.8. *Cross-region transfer protocol*

For each ordered pair of regions the model is fitted on the source population and applied to the target population with **no target label used at any point**: no refitting, no threshold selection, no calibration. Both
445 the baseline and the thermal feature set are transferred, so that the thermal block’s contribution is a paired difference within a direction rather than a comparison across directions. All twenty ordered directions among the five regions are computed.

3.9. *Label-blind domain adaptation*

450 Two label-free remedies are tested on every direction, in both feature sets.

Region-wise z-score. Each region’s numeric features are standardised using its own statistics, source statistics from source data and target statistics

from target data, never pooled. The classifier is refitted on the z-scored source and applied to the z-scored target. This removes first- and second-
455 order marginal offsets.

CORAL after region-wise z-score. Starting from the z-scored features, the source covariance is aligned to the target covariance by the standard whitening-recolouring map [37], with regularisation $\lambda = 10^{-5}$ and eigenvalues floored at 10^{-12} . Critically **the transform is applied to the source**
460 **only**; the target is left exactly as it is. The classifier is refitted on the aligned source. Neither variant sees a target label, and both are verified label-blind at run time.

λ sensitivity was assessed over nine values from 0 to 10^{-1} on four directions in both feature families, moving transfer AUC by at most 0.014 in
465 any direction and 0.008 within the thermal family. The canonical $\lambda = 1$ lies outside the released sweep and is reported separately in Appendix A(b).

3.10. Transfer-gap decomposition and the concept-shift diagnostic

For each direction the gap between the target’s own within-region skill and the raw transfer result is split in two. One part is what the best label-free
470 adaptation recovers, and the other is what it does not. The recovered fraction is defined as $(\text{adapted} - \text{raw}) / (\text{within} - \text{raw})$, signed and unclipped, and its interval comes from the same paired bootstrap. The recovered part bounds what covariate-level correction can achieve. The remainder is unrecovered by that correction; Section 4.3 shows it should not be read as a conditional
475 residual, because much of it is incurred inside a single region.

The mechanism is diagnosed by **signed univariate association**. For each numeric predictor the raw ROC-AUC of that predictor against burned

is computed in each region and never folded to $\max(\text{AUC}, 1 - \text{AUC})$, so a value below 0.5 is read as a direction rather than as weakness. A reversal is
480 called bootstrap-supported only when the two regions' point estimates fall on opposite sides of 0.5 **and each region's own interval excludes 0.5**, under the same 10-cell spatial-block bootstrap. That is stricter than requiring the two regions' intervals to be disjoint: a feature whose intervals are disjoint but one of which straddles 0.5 has not been shown to point anywhere in
485 that region, so it is recorded as a point reversal only. Appendix B states the rule again beside the counts, and `conditional_similarity_transfer.json` carries it as machine-readable metadata.

3.11. Transferability diagnostics versus transfer

Twenty candidate diagnostics from five families are computed for every
490 region pair and rank-correlated (Spearman) against observed raw transfer AUC, under one bootstrap framework that resamples unordered region pairs with both of their ordered directions travelling together. The families are marginal predictor-space measures $P(x)$, including area-of-applicability dissimilarity, climatic and geographic distance and a learned domain classifier;
495 burned-niche overlap $P(x|y=1)$; burn-pattern regime distance $P(y)$; and conditional direction agreement $P(y|x)$. No family-wise error control is claimed, and the number of variants computed within each family is reported with the result. Appendix C.2 gives the full specification of each diagnostic and of the bootstrap.

500 *3.12. Interventions*

Pooled multi-region training. Leave-one-region-out: the model is trained on the pooled primary populations of the other four regions and evaluated on the held-out region, with folds blocked as before. This asks whether pooling recovers what single-source transfer loses.

505 **Removal of direction-reversing features.** The two predictors whose signed association reverses between regions with bootstrap support are **elevation_mean** (Manavgat against Bejís and against Muğla) and **lst_anomaly_mean** (Bejís against Evia). They are dropped and everything is refitted, within-region and across every direction, and the two are also dropped singly so the
 510 cost can be attributed. This measures what the removal costs locally and what it returns on transfer on the same footing. Note that the two features are selected by the same reversal analysis against which the result is then read, so both quantities are post-selection estimates and no correction for that selection is applied.

515 *3.13. Controls on the transfer path*

Four evaluations establish what the transfer arms are measuring. All use the transfer protocol of Section 3.8 unchanged: fit on the source cells, apply to the target cells, no refit, no recalibration, no threshold selection. All were verified to reproduce the frozen cross-region AUCs exactly before being used.

520 **Within-region half-split.** The modelled cells of a region are cut in two at the median of a grid axis, a model is fitted on one half and applied to the other. Both axes and both directions are run, giving four splits per region. A split is discarded when either half is single-class, which happens twice in

one region. Per-split source and target positive counts are reported, because
525 they are unequal and bear on the interpretation.

Foreign-region evaluation. Each held-out scar area is additionally
scored with a model fitted on each of the other four regions in turn, and
those four results averaged, so that a same-region model and a foreign one
are compared on identical cells. This is the arm that separates the effect of
530 withholding the fire from the effect of separation distance.

Leave-one-scar-out. Burned connected components are found with 8-
connectivity on the analysis grid. Every component of at least 50 cells is
held out together with all cells within a buffer of it, a model is fitted on the
remainder of the same region, and applied to the held-out area. Buffers of
535 2, 5 and 10 km are run. A scar held out at 2 km inside its own region is
close to its training data and yet wholly unseen, which is what makes the
comparison with the foreign-region arm informative. The held-out patch is
defined by the labels, so this arm cannot separate the identity of a fire from
its location.

540 Appendix A(i) reports all four in full, with per-split and per-scar tables.
Intervals for the four-row ladder are Student t over the eight held-out scars,
which is that arm’s resampling unit, rather than the spatial-block bootstrap
of Section 3.7 used everywhere else in this paper.

3.14. *Leakage control*

545 An explicit forbidden-column set is enforced at every model fit. Coordi-
nates (`lon`, `lat`, `row`, `col` and their normalised forms), every burn-date and
label-provenance column, and the agreement fraction are excluded from all
feature sets, and the enforcement runs as an assertion rather than a con-

vention. The natural-vegetation mask is used only to define the population,
550 never as a predictor. Spatial blocking prevents a cell from sharing a fold with
its own neighbours.

Reproducibility. All randomness uses seed 42 and the bootstrap uses
1000 replicates throughout. The transfer and adaptation analysis runs in
an environment separate from the upstream pipeline’s. Every within-region
555 model was therefore refitted there and compared against the frozen up-
stream output, and the independently implemented adaptation was com-
pared against the pipeline’s own. The within-region comparisons agree ex-
actly, and the twenty transfer directions agree to within 1.6×10^{-7} . Random-
forest fits are not bit-identical across library versions. All numbers here were
560 therefore produced under scikit-learn 1.9.0, or verified against it. If that
version is not fixed, cross-region point estimates carry an implementation
tolerance of about ± 0.02 to 0.03 . The companion paper reports the version
sensitivity and the reproduction check in full.

Sensitivity analyses. Every headline result is repeated across two anal-
565 ysis populations, three spatial-block sizes, the CORAL regularisation sweep
and both feature sets. Where a conclusion depends on one of those choices,
the dependence is reported rather than resolved by choosing the favourable
setting.

3.15. Same-geography event-to-event comparison (Muğla 2021 versus 2022)

570 Muğla admits a comparison in which place is held fixed and the event
varies: a second fire burned inside the identical AOI, on the identical analysis
grid, eleven months after the first. Signed univariate AUCs are computed
for both arms under the same 10-cell spatial-block bootstrap used elsewhere.

Season, year and population all differ between the arms, since the 2022 arm
575 is defined by removing the 2021 scar, and Section 4.8 reports what that costs.
Appendix C.3 gives the window dates, the population construction and the
asymmetry audit.

4. Results

4.1. Study regions and the admissibility gate

580 All five candidate regions pass the burned-landcover gate as wildfire can-
didates and the negative control fails it as intended. Kozan 2023 returns
a natural-vegetation fraction of 0.017 against the 0.50 threshold and is ex-
cluded from all modelling. The separation is not marginal: the five admitted
585 regions carry fractions of 0.723 to 0.991, so the threshold falls in an empty
interval rather than between neighbouring cases. That is consistent with the
gate separating burned area produced by natural-fuel combustion from post-
harvest stubble burning, which MCD64A1 does not distinguish, though one
negative control cannot establish it.

North Evia is analysed on an extended AOI. Relative to the legacy box,
590 the extended box leaves the burned scar essentially unchanged while cutting
overall prevalence from 0.361 to 0.122 and TSG prevalence from 0.676 to
0.287. The effect of that choice on transfer is reported in Appendix A(a).

4.2. Within-region: the thermal increment replicates in five regions

Adding the six thermal predictors to the baseline raises spatially blocked
595 out-of-fold ROC-AUC in every region. The increment's bootstrap interval
excludes zero in all five regions at 1 km and at 5 km blocking (Fig. C.3).

Table 2: Region summary: populations and gate outcomes. Counts from each region’s Step 8A dataset statistics; gate fractions from each region’s burned-landcover gate output. TSG = the primary natural-vegetation population. The TSG columns use the canonical modelled population, `burnable_tree_shrub_grass` and `valid_for_modeling == True`, which is the population every model in this paper was fitted and scored on.

Region	Total cells	Valid cells	Burned	Prevalence (all valid)	TSG cells	Burned in TSG	TSG prevalence	Burned natural- veg frac- tion	Gate verdict
Manavgat 2021	24,150	24,087	796	0.033	20,511	784	0.038	0.984	pass
Bejís 2022	15,759	15,759	1,103	0.070	15,190	1,100	0.072	0.991	pass
Muğla 2021	73,098	73,045	3,026	0.041	41,730	2,911	0.070	0.958	pass
North Evia 2021 (ex- tended)	22,925	22,906	2,788	0.122	9,298	2,664	0.287	0.945	pass
Monte Fitz 2021	3,234	3,173	697	0.220	2,544	539	0.212	0.723	pass

Those are the two scales this design supports as intervals. At 10 km the point estimates hold, from +0.048 to +0.154. They rest on 6 to 33 positive-carrying blocks, so they are reported as indicative rather than as intervals.

600 The table note gives the reason.

Table note (resampling units). The bootstrap resamples spatial blocks, so what bounds an interval’s reliability is the number of blocks that carry at least one burned cell. Those counts fall quickly as blocks coarsen. At 2 cells they are 235 (Manavgat), 302 (Bejís), 843 (Muğla), 716 (Evia) and 605 192 (Montiferru), out of 5 439, 3 967, 11 316, 2 566 and 743 blocks. At 10 cells they are 28, 19, 70, 41 and 16, out of 237, 176, 576, 155 and 35. At 20 cells they are **12, 6, 33, 15 and 6**, out of 60, 48, 167, 50 and 12. No bootstrap replicate was invalid at any block size, with one exception. At 20 cells Bejís had 5 of 1000 replicates single-class. That is the symptom 610 of the six positive-carrying blocks just reported, and a further reason to read the row as indicative. An equal-tailed percentile interval built on six positive-carrying blocks has no meaningful coverage. Montiferru at 20 cells also feeds only 12 groups into a 5-fold grouped split, so its models train on about ten blocks each. **The 20-cell row of this table should be read** 615 **as indicative rather than as an interval.** The 10-cell row is the coarsest blocking this design supports properly, because every region there has 16 to 70 positive-carrying blocks. The increment holds at that scale in all five regions. Source: `paper/referee2_numbers.md`, block C, counted from the frozen per-cell prediction tables.

620 This within-region result is not itself novel. It is reported because the transfer arms below are measured against it.

Table 3: Within-region baseline versus thermal performance and block-size robustness. Primary (TSG) population; spatially blocked 5-fold CV (Section 3.7); paired spatial-block bootstrap, 1000 replicates. Block sizes 2/10/20 cells $\approx$ 1/5/10 km.

Region	Block ($\approx$ scale)	Baseline AUC	Thermal AUC	Δ AUC	Δ AUC 95% CI
Manavgat 2021	2 ($\sim$ 1 km)	0.803	0.870	+0.067	[+0.055, +0.079]
	10 ($\sim$ 5 km)	0.748	0.797	+0.050	[+0.023, +0.077]
	20 ($\sim$ 10 km)	0.683	0.731	+0.048	[+0.014, +0.085]
Bejís 2022	2 ($\sim$ 1 km)	0.862	0.918	+0.056	[+0.048, +0.065]
	10 ($\sim$ 5 km)	0.779	0.825	+0.045	[+0.018, +0.069]
	20 ($\sim$ 10 km)	0.739	0.795	+0.057	[+0.031, +0.090]
Muğla 2021	2 ($\sim$ 1 km)	0.743	0.859	+0.116	[+0.106, +0.125]
	10 ($\sim$ 5 km)	0.698	0.777	+0.079	[+0.050, +0.105]
	20 ($\sim$ 10 km)	0.673	0.733	+0.061	[+0.030, +0.094]
North Evia 2021 (ext.)	2 ($\sim$ 1 km)	0.759	0.912	+0.153	[+0.142, +0.166]
	10 ($\sim$ 5 km)	0.716	0.864	+0.148	[+0.119, +0.183]
	20 ($\sim$ 10 km)	0.679	0.833	+0.154	[+0.124, +0.189]
Montiferu 2021	2 ($\sim$ 1 km)	0.781	0.883	+0.101	[+0.080, +0.125]
	10 ($\sim$ 5 km)	0.620	0.720	+0.100	[+0.017, +0.186]
	20 ($\sim$ 10 km)	0.555	0.681	+0.126	[+0.053, +0.228]

4.3. Cross-region transfer, and what label-free adaptation does to it

Raw transfer is heterogeneous and includes anti-predictive directions. Raw target AUC spans 0.326 to 0.686 (Fig. C.4). Twelve of 20
625 directions are above chance with CI support. Six are *below* chance with CI support: both directions of Manavgat to Bejís and of Manavgat to Muğla, plus Bejís to Evia and Evia to Bejís. Two intervals span 0.5. Even the best raw transfer, Evia to Manavgat at 0.686, stays far below that target’s own within-region thermal performance of 0.870. Across all directions the raw
630 deficit against the within-region reference is 0.184 to 0.592 AUC. That reference is a region-level blocked estimate; the controls below show it is not matched to a transfer evaluation.

Those counts belong to the 2-cell blocking of Table 7. At the more conservative 10-cell blocking the same points give 9 above, 4 below and 7 uncertain,
635 and no direction changes side of the chance line (Section 4.7). Four of the six below-chance directions keep their support there. Bejís to Manavgat and Manavgat to Muğla lose it and carry no verdict. The qualitative statement is unchanged. The counts should not be read as exact.

In precision terms the transfer is worse than the ROC figures suggest. ROC-AUC is the metric used throughout this paper, for comparability with the literature, but a susceptibility surface is used as a ranked area budget, so precision-recall is the operational quantity. Across the twenty directions the thermal model’s PR-AUC averages **0.156 against a no-skill baseline of 0.136**. The mean of the twenty per-direction lifts is 1.16; the
645 ratio of the two means just quoted is 1.146. **Six of the twenty fall below their own no-skill baseline at the point estimate, and five of those**

six have intervals entirely below it. The exception is Bejís to Manavgat, at 0.034 [0.029, 0.042] against a baseline of 0.038, whose interval covers the baseline and which is therefore a point-estimate case only. Only one direction, Evia to Manavgat, exceeds twice its baseline, at 0.094 against 0.038. A transferred model therefore ranks burned cells about a sixth better than random on average, and worse than random in five directions with interval support and a sixth at the point. This is a sharper statement than the ROC means support and it should be the one a practitioner reads.

The static baseline does not transfer either. This is the control for the reading the rest of the paper invites, and it constrains that reading sharply. The same twenty directions were run with the terrain, fuel and greenness baseline alone. The mean target AUC is **0.537**, against **0.541** for the thermal model. Static attributes of a place are the class Dimarco et al. transfer successfully, and are the class this paper’s framing treats as portable. Here that class is itself barely above chance. The paired per-direction contrast is given in Section 4.6b. What matters here is that the transfer failure below is not specific to the dynamic block. A baseline that does not travel, plus pre-fire thermal state, gives a model that does not travel.

The thermal block’s paired contribution to transfer, with its interval. The two matrices were differenced direction by direction. The mean is **+0.004**, but the mean is not the informative statistic here. The individual paired contributions span **−0.148 to +0.133**, with **twelve positive and eight negative** (`baseline_vs_thermal_transfer.csv`). The spread is thirty times the mean, and the sign is not a property of the block but of the pair it is asked to cross: the same six predictors that add +0.133 in one

direction subtract 0.148 in another. A mean near zero here records cancel-
 lation, not consistent absence of effect. The directions are not independent,
 because each region appears in eight of the twenty. The interval therefore
 675 depends on what is treated as the resampling unit. All four units the design
 permits give the same answer:

Table 4:

Resampling unit	n	95 % interval on the mean paired contribution
Directions, naive	20	$[-0.027, +0.034]$
Unordered pairs, cluster bootstrap	10	$[-0.028, +0.036]$
Unordered pairs, t on pair means	10	$[-0.034, +0.042]$
Regions, leave-one-out jackknife	5	$[-0.037, +0.046]$

Every interval spans zero, and the point estimate is a small fraction of
 each width. The per-region jackknife shows how little the mean is anchored.
 Holding out Manavgat, Bejís, Muğla, Evia and Montiferru in turn gives
 680 $+0.0148, +0.0021, +0.0065, -\mathbf{0.0081}$ and $+0.0060$. **Dropping Evia alone
 reverses the sign of the headline.** None of these four units propagates
 within-direction sampling variability. They resample between directions only.

Where the skill is lost, on a matched comparison. Four evaluations
 are reported. The last three are scored on **identical cells**, so they differ only
 685 in what the model was trained on, and the first is included to show why an
 unmatched comparison misleads. Each uses the transfer protocol unchanged:
 fit, then apply with no refit, no recalibration and no threshold selection.

The held-out unit is a burned connected component of at least 50 cells
 together with all cells within 2 km of it. What makes that a harder discrim-

690 ination problem than a whole region is the composition of its negatives, not
its burned fraction, and that can be shown rather than argued. Taking the
same fitted model and the same out-of-fold predictions, and scoring them on
a random sample of region cells drawn at the scar area’s own burned frac-
tion, gives **0.782 against the region-wide 0.782**: matching the prevalence
695 changes nothing, at $-0.000 [-0.003, +0.002]$ over the nine scars. Scoring the
same predictions on the scar area itself gives 0.627, a fall of **+0.155 [+0.093,
+0.217]**. The whole effect is therefore the negative pool. Every negative
in a scar collar is fire-adjacent, sharing the terrain, land cover and synoptic
conditions of the positives, whereas a region’s negatives include its easy far
700 field. The burned fraction, 34 to 87 % against 3.8 to 28.7 % for a region, is
a symptom of that construction, and ROC-AUC is in any case invariant to
class balance at fixed class-conditional distributions.

Table 5: The four evaluations, scored on identical cells. Primary natural-vegetation pop-
ulation. Rows B, C and D are scored on the held-out scar area; row A is the whole region
and is shown to make the mismatch visible. Means and Student t intervals are over the
eight held-out scars.

Evaluation	Model trained on	Scored on	Mean AUC	95 % CI
A. Blocked cross-validation, 5 km	the region, scar included	the whole region	0.776	[0.738, 0.814]
B. Same blocked model, restricted	the region, scar included	the scar area	0.634	[0.552, 0.716]
C. Leave-one- scar-out	the region, scar withheld	the scar area	0.552	[0.501, 0.602]
D. Foreign region	another region, 306 to 2,802 km	the scar area	0.555	[0.495, 0.616]

All four rows are means over the **same eight scars**. A ninth burned component, Bejís, is excluded throughout: it is that region's only component
705 of any size, so holding it out leaves no usable source model and there is no row C for it. Rows A, B and D are reported here on the eight so that the differences are paired, and the intervals are Student t over those eight, which is the resampling unit for this arm rather than the spatial-block bootstrap used elsewhere in the paper. Eight is a small number and the intervals are
710 wide accordingly. Four of the eight scars are in Muğla and two in Montiferru, so they are not independent; row A in particular is a region-level quantity repeated across the scars of a region, and its interval is pseudo-replicated and should not be read as coverage. The two differences are computed per scar and paired, which is what they are reported for.

715 **The same model, scored two ways on the same region, differs by 0.143 AUC.** Rows A and B are one model. The only change is which cells it is scored on: the whole region, or the burn scar and its 2 km collar. That change alone costs **0.143** [**+0.077**, **+0.208**] of the 0.225 [+0.157, +0.293] fall from A to C, about two thirds, and it is the size of the entire increment
720 this literature usually reports. Both intervals are Student t over the eight scars, so the "two thirds" is a ratio of two estimates and is compatible with anything from roughly a third to nine tenths; the point of the comparison is the size of the numerator, not the precision of the fraction. A region-wide blocked figure is therefore an upper bound on what the same model achieves
725 where the fire actually is, and the difference is not small enough to ignore. Any comparison of a scar-level result against a region-level reference inherits that, and the region-level reference is what a paper of this kind normally

reports.

Neither withholding the fire nor moving 2,800 km has a measurable cost. B minus C, the fire-specific residual on identical cells, is **+0.082** 730 **[−0.011, +0.175]**, which spans zero. C minus D, the effect of replacing a same-region model with one fitted between 306 and 2,802 km away, is **−0.003** **[−0.075, +0.069]**, which also spans zero and is an order of magnitude smaller. The second is the cleaner result: on the same cells, a model 735 fitted in another country does as well as one fitted in the same region with the scar withheld, and it survives restriction to Muğla, the only region where holding out one scar still leaves the source model properly trained, at 0.579 against 0.597.

Row D is a mean over four foreign models, and the spread behind it is large. Decomposing it gives **32** scar-by-source combinations, 740 eight scars by four foreign sources, running from 0.374 to 0.724, with **ten of the 32 below chance** and a mean spread of 0.171 across the four sources for a given scar, reaching 0.329 for one. The pooled 0.555 is therefore not a statement that any foreign model does as well as a same-region one. It is a 745 statement that the *average* foreign model does, and the variation it averages over is the pair-specific sign instability that Sections 4.4 to 4.8 are about. Both facts belong together: a region boundary costs nothing on average, and which side of that average a particular pair lands on is not predictable from anything measured here.

750 The first difference should not be over-read, and one confound has to be stated. Holding out a region’s only large scar also removes most of its positives: the source model retains 11 positives in Evia, 88 in Manavgat

and 97 in Montiferru, against about 2,000 in Muğla, which has four separate scars. Those four non-Muğla arms average 0.525 and the four Muğla arms
755 0.579. Three of the four are genuinely starved; the fourth, Montiferru's second component, retains 472 source positives, so the shortfall there is not a training-size effect. so row C mixes "the fire was withheld" with "almost all the positives were withheld". Restricted to Muğla, C is 0.579 and D is 0.597, and the C to D comparison still shows nothing. With eight scars, three of
760 them starved, this design cannot establish a fire-specific residual, only bound it at about 0.18 against the 0.143 that separates row A from row B on the same model.

What this establishes is bounded rather than positive. Rows C and D both sit close to chance, so a comparison between them has little dynamic
765 range, and neither the fire's identity nor the region boundary is shown to cost anything on this evaluation. What is measurable is the change of evaluation geometry between rows A and B. What it does not establish is the fire event as the unit, because the held-out patch is defined by the labels, so its identity cannot be separated from its location. Two further observations
770 point the same way: a model trained on the other half of the same fire is no better at matched sample size, and the within-region half-split, which does see half of the target scar, returns 0.574 against leave-one-scar-out's 0.552 with overlapping intervals.

Separation does not order the twenty-direction matrix either. The Spear-
775 man correlation between centroid separation and transfer AUC is -0.32 with an interval spanning zero, and the two nearest directions, Manavgat and Muğla at 306 km, are among the worst at 0.470 and 0.401. This is a different

quantity from Table 9’s geographic-distance row, computed on the twelve-direction subset, which gives -0.24 ; both span zero. The transfer mean of
780 0.541 is over all twenty directions and is not the value at any one separation: the 2,802 km pair returns 0.326 and 0.444.

The result is robust to how the held-out patch is defined, at 0.543 to 0.565 across minimum component sizes from 25 to 200 cells and under both 4- and 8-connectivity, and flat across buffers of 2, 5 and 10 km.

785 **The increment declines with the holdout, and is not established once the fire is withheld.** Contribution 1 is about the paired thermal-minus-baseline difference, so the same evaluations were run on it.

Table 6:

Evaluation	Thermal increment	Interval
Blocked cross-validation, per region	+0.056 to +0.153	every interval above zero
Within-region half-split	+0.027	positive in 13 of 18 splits
Leave-one-scar-out	+0.022	[−0.032, +0.077]
Cross-region, twenty directions	+0.004	[−0.028, +0.036]

The point estimate falls monotonically as the holdout hardens, and the last two intervals span zero. Under leave-one-scar-out the increment is posi-
790 tive in six of eight scars and averages +0.022, against a baseline of 0.529 and a thermal 0.552 that are both close to chance on that evaluation; restricted to Muğla it is +0.009 [−0.051, +0.070]. **The within-region increment of +0.056 to +0.153 is therefore substantially a property of interleaved holdout.** It is not shown to be zero on an unseen fire, and eight scars

795 cannot show that; what this design establishes is that it is not established there.

On the frames as drawn, six directions are below chance with interval support, the sharpest at 0.326 [0.305, 0.349]. No account of merely lost skill produces a reliably reversed ranking, so on this evidence those six
800 would need a mechanism acting on the direction of the relationship, and Sections 4.4 to 4.8 pursue one. **Section 4.9 then shows that most of this count is an artefact of the evaluation frames: equalising them leaves one direction below chance, not six.** The reader should carry that forward through the intervening sections, whose reversal evidence is computed
805 on the frames as drawn. Per-split and per-scar detail is in Appendix A(i).

Label-blind adaptation compresses the matrix toward chance rather than repairing it. Under region-wise z-scoring the twenty directions span 0.431 to 0.630 and under CORAL 0.443 to 0.624, roughly half the raw spread, with no adapted direction exceeding 0.631 against the un-
810 matched within-region references of 0.859 to 0.918. Adaptation raises the failing directions and degrades most of those that already transferred. Taking the better of the two adaptations per direction, 14 of the 20 end closer to chance than they began and 6 end further from it; five of those six involve Montiferru, the smallest and last-added region, and move upward, while the
815 sixth is Manavgat→Muğla moving downward from 0.470 to 0.443. The 14 to 6 split should be read at the precision of limitation (ix) in Section 5.9, since Bejís→Manavgat is counted as compressed on a margin of 0.001.

Against the right reference, adaptation is not failing (Fig. C.5). The three controls above give an achievable reference for a model applied to

Table 7: Cross-region transfer matrix, thermal model, TSG population. Target ROC-AUC with 2-cell spatial-block bootstrap 95% CIs (1000 replicates). CORAL is applied after region-wise z-scoring ($\lambda = 10^{-5}$).

Direction	Raw	Region-wise z-score	CORAL
Manavgat→Bejís	0.326 [0.305, 0.349]	0.477 [0.451, 0.502]	0.511 [0.484, 0.534]
Bejís→Manavgat	0.444 [0.408, 0.480]	0.457 [0.420, 0.497]	0.555 [0.528, 0.583]
Manavgat→Muğla	0.470 [0.451, 0.488]	0.431 [0.411, 0.449]	0.443 [0.423, 0.461]
Muğla→Manavgat	0.401 [0.378, 0.426]	0.559 [0.531, 0.587]	0.560 [0.535, 0.587]
Manavgat→Evia	0.613 [0.593, 0.631]	0.542 [0.520, 0.565]	0.539 [0.518, 0.561]
Evia→Manavgat	0.686 [0.653, 0.716]	0.516 [0.489, 0.544]	0.527 [0.500, 0.553]
Bejís→Muğla	0.618 [0.601, 0.635]	0.518 [0.501, 0.535]	0.507 [0.489, 0.524]
Muğla→Bejís	0.583 [0.561, 0.607]	0.535 [0.512, 0.557]	0.560 [0.538, 0.581]
Bejís→Evia	0.383 [0.363, 0.402]	0.532 [0.509, 0.551]	0.499 [0.479, 0.518]
Evia→Bejís	0.448 [0.426, 0.470]	0.549 [0.524, 0.575]	0.549 [0.525, 0.573]
Muğla→Evia	0.653 [0.636, 0.671]	0.561 [0.543, 0.580]	0.563 [0.545, 0.582]
Evia→Muğla	0.577 [0.560, 0.593]	0.501 [0.485, 0.518]	0.530 [0.515, 0.546]
Montiferru→Manavgat	0.567 [0.539, 0.594]	0.573 [0.538, 0.609]	0.606 [0.574, 0.639]
Manavgat→Montiferru	0.533 [0.488, 0.580]	0.586 [0.540, 0.629]	0.592 [0.550, 0.631]
Montiferru→Bejís	0.548 [0.521, 0.578]	0.574 [0.552, 0.596]	0.569 [0.548, 0.591]
Bejís→Montiferru	0.594 [0.560, 0.631]	0.550 [0.500, 0.601]	0.574 [0.530, 0.621]
Montiferru→Muğla	0.619 [0.604, 0.634]	0.576 [0.562, 0.589]	0.565 [0.550, 0.579]
Muğla→Montiferru	0.531 [0.495, 0.568]	0.587 [0.549, 0.624]	0.584 [0.547, 0.623]
Montiferru→Evia	0.586 [0.565, 0.606]	0.630 [0.611, 0.649]	0.624 [0.605, 0.641]
Evia→Montiferru	0.647 [0.608, 0.682]	0.568 [0.528, 0.609]	0.581 [0.539, 0.623]

Table 8: Transfer-gap decomposition (four-AOI set, 12 directions). Within = target’s within-region thermal AUC; best adapted = the better of z-score/CORAL; recovered fraction = (adapted – raw)/(within – raw), signed and unclipped, with paired bootstrap CI (1000 replicates). Montiferru directions are not part of this decomposition (per-pair absolute decompositions exist without fraction CIs). The status column asks whether the *adapted* value clears chance and uses the 2-cell adapted intervals of Table 7. The adapted arms were not recomputed at the coarser blocking of Appendix A(c), which covers the raw arm and the paired delta only.

Direction	Within	Raw	Best adapted (method)	Recovered fraction [CI]	Status
Manavgat→Bejís	0.918	0.326	0.511 (CORAL)	+0.31 [+0.28, +0.34]	recovery, chance not excluded
Bejís→Manavgat	0.870	0.444	0.555 (CORAL)	+0.26 [+0.18, +0.34]	recovery above chance
Muğla→Manavgat	0.870	0.401	0.560 (CORAL)	+0.34 [+0.28, +0.40]	recovery above chance
Bejís→Evia	0.912	0.383	0.532 (z-score)	+0.28 [+0.23, +0.32]	recovery above chance
Evia→Bejís	0.918	0.448	0.549 (z-score)	+0.22 [+0.16, +0.27]	recovery above chance
Manavgat→Muğla	0.859	0.470	0.443 (CORAL)	−0.07 [−0.12, −0.03]	negative recovery
Muğla→Bejís	0.918	0.583	0.560 (CORAL)	−0.07 [−0.16, +0.01]	negative recovery
Manavgat→Evia	0.912	0.613	0.542 (z-score)	−0.23 [−0.32, −0.14]	negative recovery
Evia→Manavgat	0.870	0.686	0.527 (CORAL)	−0.86 [−1.18, −0.60]	negative recovery
Evia→Muğla	0.859	0.577	0.530 (CORAL)	−0.17 [−0.22, −0.11]	negative recovery
Muğla→Evia	0.912	0.653	0.563 (CORAL)	−0.35 [−0.43, −0.27]	negative recovery
Bejís→Muğla	0.859	0.618	0.518 (z-score)	−0.42 [−0.51, −0.34]	negative recovery

820 a fire it has not seen: 0.574 for the half-split, 0.552 for the leave-one-scar-out. CORAL, committed to in advance, averages **0.552** across the twenty directions against 0.541 raw, which is exactly that reference. Taking whichever of the two adaptations scores higher per direction gives 0.556, but that selection uses the target labels the protocol forbids, so it is an oracle upper
825 bound rather than an achievable result and is reported as one. What it does is regress the matrix towards that value: fourteen of twenty directions move closer to chance, which harms the directions that already worked and helps the ones that did not. The verdict that follows is narrower than "alignment fails". Alignment cannot exceed what a model can achieve on an unseen fire,
830 and it does not.

In the six directions where raw transfer was below chance, the best label-free method recovers at most 34 % of the gap to the within-region reference, so the remaining unrecovered fraction is at least 0.66 everywhere. The half-split control above shows that the larger part of that remainder is already
835 incurred inside the region, so it should not be read as a measure of concept shift. Seven directions show *negative* recovery, meaning adaptation moves the score away from the reference; in six of those raw transfer was already above chance and adaptation destroyed that advantage. Label-free alignment therefore does not act as a repair mechanism.

840 4.4. *Transferability diagnostics: only conditional similarity orders transfer*

Twenty candidate diagnostics from five families were each rank-correlated with the same target quantity, the raw thermal transfer AUC over the twenty ordered directions, under one common pair-based bootstrap.

Only two diagnostics have intervals excluding zero, and both

Table 9: Transferability diagnostics versus raw thermal transfer, by family. Spearman ρ against raw transfer AUC with pair-based bootstrap 95 % CIs. Exp. is the sign expected if the diagnostic orders transfer. The member named is the one with the largest absolute correlation in its family, which is not always in the expected direction. All twenty individual diagnostics are in Appendix B, Table B1.

Family	Diagnostics	Exp.	Largest correlation in family	Sign as expected	Any CI excluding 0
P(y x) condi- tional	6	+	+0.84 [+0.58 , +0.88] agree- ment fraction, supported features	yes	yes, 2 of 6
P(x y=1) niche overlap	5	+	+0.24 [−0.45, +0.74] Schoener’s D, 1-D mean	yes	no
P(x) marginal	6	−	−0.32 [−0.78, +0.33] domain- classifier AUC	yes	no
P(y) regime structure	2	−	+0.29 [−0.38, +0.74] ^{log} ₄₃ effective- N distance	no	no
geographic	1	−	−0.24 [−0.84,	yes	no

845 **are conditional.** The stronger is the sign-agreement fraction over interval-supported features, at $\rho = +0.84$ $[+0.58, +0.88]$. The cosine variant reaches $+0.81$. No marginal measure was shown to order the matrix. That includes area-of-applicability-style dissimilarity in predictor space, climatic distance and geographic distance. The niche-overlap and regime families were not
850 shown to order it either. The learned domain classifier is at ceiling, separating source from target at $\text{AUC} \geq 0.96$ for every pair. It always succeeds, which is why it carries no ordering information.

Two limits are stated with the result rather than after it. The first is size. The index’s tie structure caps the achievable Spearman at $+0.861$,
855 so the observed $+0.840$ sits essentially on that ceiling. Its exact one-sided permutation p is 0.0060 , the smallest this tie structure can produce, against a Bonferroni threshold of 0.0026 over the nineteen computed variants. No outcome of this diagnostic could have cleared family-wise correction on ten effective pairs. The second limit is labels. Signed associations need burned
860 labels in both regions. The family that works is therefore not available before deployment, while the family that fails is.

A third limit is about selection, and it is the sharpest of the three. The two rows that clear zero are the *supported-feature* variants. Their feature subset is chosen by whether two regions’ bootstrap intervals happen to be
865 disjoint. That is a data-dependent selection, made on the same data, with no correction. The unselected counterparts over all nine features are $\rho = +0.50$ $[-0.17, +0.83]$ for the cosine and $\rho = +0.18$ $[-0.40, +0.72]$ for the agreement count. Both span zero. The result lives in the selection step, and is reported as such.

870 **The families are compared on unequal samples, and equalis-**
ing them does not change the ordering. The marginal, applicability,
climatic and geographic rows sit on twelve directions, because those diag-
nostics exist only for the four-region subset. The supported-conditional rows
sit on sixteen and the rest on twenty. A reader may reasonably ask whether
875 the marginal family’s failure is a statement about power rather than about
diagnostics. Every row was therefore recomputed on the common twelve di-
rections. The published values reproduce to 4.8×10^{-5} . The conditional
rows still lead, at +0.87 [+0.65, +0.88] and +0.85 [+0.43, +0.88], and every
marginal row still spans zero. The ordering is not an artefact of unequal
880 samples.

4.5. *The contrast pair: similarity is not sufficient*

The clearest single view of Table 9 needs no ranking at all (Fig. C.8).
Manavgat and Muğla lie in the same country and the same fire year. They
are 306 km apart by the centroid geodesic distance this paper uses as a
885 diagnostic, and their nearest boundaries are 191 km apart. Their burned cells
occupy the most similar environmental envelope of any pair in the matrix,
with per-feature Schoener’s D of 0.77 to 0.89. Yet on the frames as drawn
five of nine feature-response directions point opposite ways. Elevation is one
of them, at a signed AUC of 0.374 [0.289, 0.471] in Manavgat against 0.611
890 [0.532, 0.690] in Muğla, with disjoint intervals, and all four absolute thermal
channels are among the others. **Section 4.9 withdraws that reading:** this
is the pair whose Manavgat arm is 60 % far field, and under a 10 km collar
the elevation figure moves to 0.561 against 0.606, on the same side of 0.5.
The contrast that survives is the transfer result itself, not the reversal count

895 that was offered to explain it. Transfer is below chance in both directions at the point estimate, at 0.470 and 0.401. Both directions also sit inside the nominal area of applicability, at 0.875 and 0.531 of target cells.

Bejís and Montiferru sit at the opposite extreme. Their burned envelopes barely overlap, and they carry the most dissimilar values on every overlap
900 measure. They transfer above chance in both directions. That half of the contrast rests on point estimates: neither direction carries a verdict at 5 km blocking, and the marginal applicability audit was never produced for Montiferru. The claim is one of *sufficiency*. Similarity does not guarantee transfer, and dissimilarity does not preclude it. The claim rests on two
905 coexisting counterexamples, so it does not depend on the number of pairs available.

4.6. Interventions: pooling and feature removal

(a) **Pooled multi-region training** (Fig. C.6). At the point estimate, training on the pooled primary populations of the other four regions never
910 beats the best single-source transfer for any target, with shortfalls of 0.02 to 0.22, and it stays 0.28 to 0.50 AUC below the within-region ceiling; for two targets it falls below the pairwise mean and below chance, though only Bejís is below chance with interval support, at 0.417 [0.369, 0.467]. Aggregation does not recover what single-source transfer loses.

915 (b) **Removing the direction-reversing features.** The two predictors whose signed association reverses between regions **with bootstrap support** are **elevation_mean** and **lst_anomaly_mean**. The per-region signed associations and the three pair-level reversals that meet that criterion are in Appendix B, Tables B2 and B3. Retraining without them costs **−0.081**

920 of mean within-region AUC, supported in every region (per-region deltas
 -0.060 , -0.130 , -0.074 , -0.063 , -0.079 , every interval entirely below zero),
and changes mean transfer by **+0.014** [**-0.017**, **+0.045**], which spans zero.

Two qualifications belong with those numbers rather than after them.
First, the debit is not the thermal block's. Dropping elevation alone accounts
925 for -0.061 of it, from 0.888 to 0.827. Dropping the LST anomaly alone
accounts for -0.013 , from 0.888 to 0.875. Roughly three quarters of the
cost is therefore the removal of a **baseline** terrain variable, not of a thermal
channel. Second, both figures are post-selection. The two predictors were
chosen because they reverse, using the same data on which the -0.081 and
930 the $+0.014$ are then estimated. No correction for that selection is applied.

Table 10:

Configuration	Mean within-region AUC	Mean transfer AUC
full	0.888	0.541
drop <code>elevation_mean</code>	0.827	0.546
drop <code>lst_anomaly_mean</code>	0.875	0.544
drop both	0.807	0.556

Fig. C.7 shows the four configurations together. What this measures is
a local cost with no compensating transfer gain. It does not measure an
exchange, because the transfer side is a null on both arms. The thermal
block's own contribution to transfer is $+0.004$, with an interval spanning
935 zero (Section 4.3). Removing the reversing predictors returns $+0.014$, with
an interval spanning zero. Two nulls on the portability axis are not a price
paid.

4.7. Sensitivity analyses

Eight design choices were varied with everything else held fixed. They
940 are the Evia AOI and its prevalence, the CORAL regularisation constant,
the blocking scale, the closure date of the predictor window, the quality
screening of the coarse thermal input, the contrast between the normalised
and the absolute dryness channels, the removal of the coordinate-informed
channels, and the capacity of the classifier. None changes a conclusion above.
945 Two bound how the results should be read, so they are carried into the main
text here.

Coarsening the blocks from 1 km to 5 km moves the **paired thermal-**
minus-baseline delta verdicts from ten positive, seven negative and three
uncertain to six, four and ten. The above-chance verdicts on the thermal
950 arm itself, a different quantity, move from twelve, six and two to nine, four
and seven. Support is removed from seven verdicts and added to none. The
point estimates are unchanged, but that is an identity rather than a result.
The blocking scale is the resampling unit, and it cannot move an estimate
computed once over all target cells.

955 Manavgat’s whole downstream chain was then rebuilt from a quality-
screened MODIS input. That changes the downscaled surface on 22,304 of
24,150 cells, by up to 10.9 °C. No signed univariate association moves by
more than +0.0003. This closes the one processing-artefact candidate for
that region’s behaviour. Appendix A reports all seven arms, including one
960 that tests a claim of Section 1.2 and does not uphold it.

4.8. *The same geography, a second fire: reversal with place held constant*

Every result above compares different places. Muğla admits a stricter test, because a second fire burned inside the identical AOI on the identical analysis grid. The 2021 event has a predictor window closing on 28 July. The 2022
965 event has a matched window closing on 20 June. Region, bounding box, cell definition, feature registry and processing chain are the same in both arms. The design holds place fixed. It does not hold season fixed, because the 2022 fire ignites about six weeks earlier, and it does not hold the population fixed, because the 2022 arm is defined by removing the 2021 scar.

970 **Elevation reverses with bootstrap support.** In 2021 higher ground burned preferentially, at 0.611 with its interval entirely above 0.5; in 2022 lower ground did, at 0.296 with its interval entirely below. The two intervals are disjoint, 0.532 against 0.355, and the difference is -0.317 $[-0.414, -0.220]$. The same predictor, the same region, the same grid, and an asso-
975 ciation that points the opposite way in two fires eleven months apart. The four absolute thermal channels change side as well, from bootstrap-supported lower-values-burn in 2021 to higher-values-burn point estimates in 2022.

Two structural properties must be read alongside these numbers, because they have no analogue in the twenty-direction matrix. The two arms are
980 not disjoint samples: they share 38,789 of the 2022 arm's 38,790 cells, and elevation, slope and land cover are identical to the digit across all 73,098 grid cells, so only NDVI and the six thermal channels carry new information between them. And the removal is the target's own positive class, so in the 2022 to 2021 direction not one target positive is present in the source training
985 population while 38,789 of 38,819 target negatives are, and membership of

Table 11: Signed univariate feature-burned AUC, Muğla 2021 versus 2022. Raw AUC against **burned**, never folded to $\max(\text{AUC}, 1 - \text{AUC})$; 10-cell (≈ 5 km) spatial-block bootstrap, 1,000 replicates, seed 42 (Section 3.15). Analysis population 41,730 rows / 2,911 burned (2021) and 38,790 rows / 331 burned (2022). **Positive-carrying 5 km blocks: 70 for the 2021 arm and 11 for the 2022 arm.** Table 3’s note sets sixteen as the floor this design supports at that blocking, so the 2022 intervals here fall below the paper’s own standard and are read as indicative, exactly as the 20-cell row of Table 3 is. The 2022 arm is additionally a single compact scar, so its eleven blocks are contiguous. Read from `paper/step9g_raw/.../mugla_2021__mugla_2022_event_relative/step9g_direction_reversal_table.csv`.

Feature	2021 AUC [95 % CI]	2022 AUC [95 % CI]	Reversal
elevation_mean	0.611 [0.532, 0.690]	0.296 [0.230, 0.355]	bootstrap- supported
current_lst_mean	0.325 [0.271, 0.382]	0.515 [0.434, 0.580]	point only
current_tvdi_mean	0.336 [0.275, 0.398]	0.594 [0.475, 0.674]	point only
downscaled_lst_mean	0.307 [0.253, 0.366]	0.508 [0.435, 0.571]	point only
fused_lst_mean	0.325 [0.272, 0.383]	0.519 [0.436, 0.583]	point only
ndvi_mean	0.662 [0.616, 0.704]	0.707 [0.624, 0.777]	none
slope_mean	0.637 [0.582, 0.686]	0.558 [0.468, 0.634]	none
lst_anomaly_mean	0.485 [0.395, 0.566]	0.380 [0.249, 0.502]	none
tvdi_difference_mean	0.490 [0.396, 0.575]	0.397 [0.265, 0.514]	none

the source training set alone separates the 2021 target’s classes at ROC-AUC 0.9996. We therefore cannot bound what that asymmetry does to a transfer estimate between the two arms, which is why none is reported here, only state that a direction with this structure is not comparable to the
 990 twenty between-region directions. The elevation reversal is the finding; the structural asymmetry is a competing explanation this design cannot exclude. The known part of the bias runs the safe way: the removed cells are high, with a median of 563 m, and unburned in 2022, so removing them raises the 2022 elevation AUC and makes the reversal smaller rather than larger.

995 *4.9. The evaluation frame, applied to our own matrix*

Section 4.3 established that swapping a region’s negatives for fire-adjacent ones moves ROC-AUC by +0.155 [+0.093, +0.217] with the model held fixed. That result was obtained inside one region. It applies with equal force between regions, and applying it changes what Sections 4.3 and 4.5 can claim.
 1000 This section reports that test. Source: `aoi_frame_auc.csv`, `aoi_frame_transfer.csv`, `paper/code/verify_aoi_frame.py`, `paper/code/verify_aoi_transfer.py`.

The five areas of interest are not comparable frames. Each region is a rectangle drawn around a fire, and the rectangles differ by an order of
 1005 magnitude in how much unburnt far field they enclose.

Montiferru’s frame is fire-scale; Manavgat’s and Bejís’s are roughly three-fifths far field. The far field is not a neutral addition. In Manavgat the median elevation of modelled cells rises from 472 m within 5 km of the fire to 955 m at 10 to 20 km and 1,273 m at 20 to 50 km, against 512 m for the burned cells
 1010 themselves, in Taurus terrain that no plausible spread model would place at

Table 12:

Region	cells	burned	median distance to burned	share beyond 10 km
Manavgat	20,511	784	13.4 km	60.1 %
Bejís	15,190	1,100	13.5 km	63.1 %
Muğla	41,730	2,911	11.3 km	55.3 %
Evia	9,298	2,664	8.0 km	43.7 %
Montiferru	2,544	539	2.7 km	2.1 %

risk.

Under an equalised frame the sign reversals of Section 4.5 do not survive. Restricting every region to cells within 10 km of any burned cell drops **no positives**, only far-field negatives.

1015 On the equalised frame all five regions agree in sign on elevation, on LST and on TVDI, and both bootstrap-supported elevation reversals of Table B3 disappear. Applying this paper’s own criterion from Section 3.10, which requires each region’s own 10-cell block-bootstrap interval to exclude 0.5, **no reversal remains bootstrap-supported in the collar frame** (`collar_`
1020 `frame_bootstrap.csv`). The closest is `lst_anomaly_mean`, at 0.392 [0.324, 0.460] in Bejís against 0.584 [**0.497**, 0.669] in Evia: the point estimates fall on opposite sides but Evia’s interval includes 0.5 by 0.003, so this is a point reversal, not a supported one. We state it that way rather than quoting the point estimates alone, because the alternative would be to apply
1025 a looser standard to the arm that supersedes Table B3 than to Table B3 itself. Elevation under the collar is above 0.5 in all five regions but individually supported in only two, Muğla at 0.606 [0.525, 0.685] and Evia at 0.648 [0.550,

Table 13:

Signed univariate AUC	Manavgat	Bejís	Muğla	Evia	Montiferru	straddles 0.5
elevation, full frame (Table B2)	0.374	0.643	0.611	0.541	0.584	yes
elevation, 10 km collar	0.561	0.614	0.606	0.648	0.581	no
current LST, full frame	0.538	0.477	0.325	0.377	0.370	yes
current LST, 10 km collar	0.386	0.405	0.332	0.286	0.376	no
current TVDI, full frame	0.552	0.517	0.336	0.362	0.356	yes
current TVDI, 10 km collar	0.392	0.454	0.342	0.250	0.361	no

0.740].

The sign the five regions agree on is not the one the dryness
1030 **framing predicts.** For LST the common direction is *below* 0.5 in every region, at 0.386, 0.405, 0.332, 0.286 and 0.376: a hotter pre-fire surface is associated with **less** burning, not more, and the same holds for TVDI. This is not a lapse-rate artefact. Pooling concordance within elevation deciles leaves LST at 0.402, 0.509, 0.363, 0.327 and 0.392, still below 0.5 in four
1035 of five regions, and within NDVI deciles at 0.440, 0.486, 0.404, 0.279 and 0.368, below 0.5 in all five, despite $r(\text{LST}, \text{NDVI})$ reaching -0.92 . The likely reading is that fuel availability dominates surface dryness at this scale in this cohort: NDVI itself is the strongest single predictor under the collar, at 0.582 to 0.663 with intervals excluding 0.5 in four of five regions, so greener
1040 and therefore more fuel-rich cells burn more, while the hottest cells are the sparse, rocky ones with little to burn. Section 1.2 motivates the thermal block from moisture-stress physics; that motivation is not what the signed associations show, and Section 5.2 states the consequence.

Three further points bound how many independent reversals could have
1045 been counted. Within the collar, `fused_1st_mean` correlates with `current_1st_mean` at 0.99 to 1.00 and `downscaled_1st_mean` at 0.97 to 0.99, and `current_tvdi_mean` at 0.87 to 0.98. The four absolute channels are close to one axis, and the two anomaly channels correlate at 0.64 to 0.94, so the "five of nine directions reverse" count of Section 4.5 is a count of features,
1050 not of independent quantities; in effective dimensions it is closer to two. That is coherent rather than fortunate. The absolute thermal channels are strongly collinear with elevation, at $r = -0.695, -0.125, -0.404, -0.511$

and -0.507 for current LST across the five regions, and -0.722 to -0.298 for TVDI; the anomaly, which is differenced against each cell’s own baseline and is therefore lapse-rate-free, correlates with elevation at only $+0.036$ to $+0.256$. The channels whose reversal vanishes are the ones that were partly measuring terrain, with a proxy strength that varies fivefold between regions. The channel whose reversal survives is the one where that explanation is unavailable.

The transfer matrix moves as well. Restricting source and target to the same collar:

Table 14:

Source frame	Target frame	Mean target AUC	Above chance	Below chance	Paired thermal delta
full	full (Table 7)	0.540	14 of 20	6	+0.003
full	10 km	0.575	17 of 20	3	+0.002
10 km	full	0.571	17 of 20	3	+0.014
10 km	10 km	0.617	19 of 20	1	+0.023
5 km	5 km	0.608	18 of 20	2	+0.014

The reference arm reproduces the frozen matrix, at 0.540 against Table 7’s 0.541 and 14 of 20 exactly, so this is measuring the same quantity. The largest movers are Bejís to Evia, 0.383 to 0.602, and Bejís to Manavgat, 0.440 to 0.601.

What this does and does not change. Three claims do not survive and are withdrawn here rather than defended. The count of six anti-predictive directions becomes one. The sign reversal of elevation, LST and TVDI as the mechanism of the residual is a property of the frames, not of the predictor-burning relationship. And the paired thermal contribution is

not stable at +0.004: it rises to +0.023 once frames are equalised, so the claim that the block contributes nothing between regions is frame-dependent too. What survives is the central negative result, and it survives with room to spare: equalised transfer of 0.617 against within-region skill of about 0.87
1075 leaves most of the gap intact. Section 5.9 records the frame as a limitation of the cohort rather than of the method, since equalising it is cheap and we recommend it.

This is the paper’s own Contribution 1 turned on the paper’s own matrix. We had measured that the composition of the negative pool moves ROC-AUC
1080 by more than the effects this literature reports, and had then computed a five-region transfer matrix, and a mechanism, on frames whose fire-adjacent share ranges from 2 % to 98 %. We report it because it is the strongest available demonstration that the reporting standard of Section 5.8 is not a technicality.

1085 4.10. *What target labels cost: the recovery curve*

Everything above measures a failure; this prices it. Label-free alignment does not close the residual gap, so the missing resource is information about the target that alignment cannot synthesise, and the direct way to supply it is target labels. A frozen labelled-budget diagnostic answers how many,
1090 for three regions across all six ordered directions, using one 10-cell (~ 5 km) spatial block as the unit of labelling effort and reading recovery against a matched target-only ceiling of 0.777 to 0.824.

Thirty-two labelled blocks recover **85 to 89 % of that ceiling in three of the six directions**, two of which started below chance; 51 to 57 % in two
1095 more, the directions where the conditional gap is widest; and 30 % in the

sixth. That budget is 7 to 20 % of the target’s natural-vegetation population, which at the 2,700 to 3,000 labelled cells those blocks carry (S1.3) and this grid’s effective cell area of 0.199 to 0.208 km² (Appendix C) is roughly **540 to 620 km²**, so it is a real answer and not a cheap one. Two properties
1100 of the measurement bound it and are stated in the supplement rather than buried: at the top budget the selection pool is nearly exhausted, so the narrow upper-budget intervals reflect saturation rather than precision, and the labelled blocks are drawn from the event being predicted, which is not a resource available before that event burns. The protocol, the full curve and
1105 the limits are in Supplement S1.

Two limits belong with the number. At small budgets the same intervention damages the direction that already transfers best without any labels. And six directions in three regions cannot support a general label budget, so none is offered.

1110 **5. Discussion**

5.1. Principal findings

Three results carry this paper, each established in Section 4 and not re-tabulated here.

First, where a model is scored decides what it appears to know. Changing
1115 only the evaluation cells costs 0.143 ROC-AUC, the cause is the composition of the negative pool rather than prevalence, and applying the same correction between regions withdraws three claims we had made from the frames as drawn, including the sign reversal we had offered as the mechanism of the transfer residual (Section 4.9).

1120 Second, the thermal block’s local skill does not travel, and the frame correction does not rescue it: equalised transfer sits at 0.617 against within-region skill near 0.87, and the static baseline travels no better than the dynamic one, so this is not a peculiarity of thermal predictors.

Third, no diagnostic that can be run before deployment was shown to
1125 order the transfer matrix. The only two that do require burned labels in the target region, which is the resource a practitioner lacks, and they rest on a data-selected subset of predictors.

5.2. Why the thermal increment is real but local

A distinction has to be drawn before this section can say anything useful.
1130 The within-region increment and the transfer failure are measured at different separations, and Section 4.3 shows that most of the difference between them is already present inside a single region. What follows therefore reads the increment as local in a specific sense: it holds where the held-out cells are interleaved with training cells, and most of it is gone once they are not,
1135 before the fire or the region changes. What that does not explain is why the directions differ from each other, and in particular why six of them are anti-predictive.

The within-region increment is not an artefact to be explained away, but it is measured under a holdout that Section 4.3 shows is generous. Withholding
1140 a whole burn scar reduces it to $+0.022$ $[-0.032, +0.077]$, so what follows describes an increment that is established under interleaved validation and not established beyond it. It replicates in five independent regions, survives coarsening of the spatial blocks to ~ 5 km with its interval intact, persists in the secondary all-valid population, and survives a predictor window closed

1145 7 and 14 days earlier in all five regions. The direction of that last result is region-specific and should not be overstated: the contribution strengthens in Bejís ($0.058 \rightarrow 0.079$ at 14 days) and Muğla ($0.115 \rightarrow 0.128$), is flat in Montiferru, and **weakens monotonically in Evia** ($0.156 \rightarrow 0.149 \rightarrow 0.135$). What holds everywhere is survival, not improvement. The transfer
1150 failure is therefore not evidence that the thermal signal is spurious. It is evidence that the fitted relationship is *local*.

The natural objection is that different Mediterranean regions are simply different systems, so that a predictor meaning one thing in one place and another elsewhere is not instability but a comparison of two different systems.
1155 The two Muğla events answer it directly: region, AOI, grid, feature registry and processing chain identical, static predictors identical cell by cell, and elevation’s association with burning reversing from 0.611 [0.532, 0.690] to 0.296 [0.230, 0.355], a difference of -0.317 [-0.414 , -0.220] (Section 4.8). Holding geography fixed does not stabilise the direction of the relationship.

1160 Three caveats attach and none is dismissed. Season and year are confounded, and **that confound cannot be resolved in this study area**: the events sit 42 days apart in median burn day-of-year, neither year contains a second event at the other’s phase, and a calendar-matched arm would carry nine burned cells against a gate minimum of thirty. The population is
1165 not held fixed either, the 2022 arm being the 2021 arm with the 2021 scar removed. And the 2022 arm rests on one compact scar and eleven positive-carrying 5 km blocks. What survives all three is the narrower reading that *something* reverses the elevation-burning association with geography, grid and pipeline held fixed, which is why Section 4.8 is reported as corroboration

1170 rather than as the load-bearing evidence for the mechanism.

The sharpest supported reversal belongs to elevation, a static predictor, so instability is a property of the predictor-to-burning mapping generally rather than of thermal channels specifically. This is the same conclusion the baseline transfer arm reaches from the other direction: the instability is not
1175 the thermal block’s peculiarity.

5.3. *Why the similarity-based diagnostics fail*

A diagnostic built on distance in predictor space asks one question: do the target’s predictor values look like the training data’s? That is not the question that matters when the failure is conditional. A target region can
1180 sit well inside the training envelope while the relationship between those predictors and burning points the other way. Manavgat to Muğla is exactly that case, with 0.875 of target cells inside the weighted area of applicability and transfer below chance.

The same holds for the niche-overlap and regime families. At the point
1185 estimates, the pair with the highest burned-niche overlap in the matrix fails in both directions while the pair with the lowest transfers in both, so no monotone function of overlap can order the outcomes. The qualifier is not decorative: at the 5 km blocking this paper otherwise defends, neither Bejís↔Montiferru direction carries a verdict and Manavgat→Muğla loses its
1190 below-chance support, so this contrast is a statement about point estimates and is made as one. The domain classifier is at ceiling for every pair, which is informative in itself: the regions are trivially separable in predictor space whether or not their transfer works, so separability carries no ordering information here.

1195 5.4. *What the conditional diagnostic is, and what it is not*

The sign-agreement index reaches $\rho = +0.84$ [+0.58, +0.88] against observed transfer. Three limits are reported with it rather than after it, and the sharpest is a selection: the two rows that clear zero are supported-feature variants whose subset is chosen on the same data, and their all-nine-feature
1200 counterparts span zero. Its tie structure caps the achievable Spearman at +0.861, so the observed value sits essentially on its own ceiling and the ranking it supports is coarse. Its exact one-sided permutation p is 0.0060, the smallest that tie structure can produce, against a Bonferroni threshold of 0.0026 over nineteen computed variants. No outcome could have cleared
1205 family-wise correction on ten effective pairs. And it needs burned labels in both regions, which makes it a mechanism diagnosis rather than a pre-deployment screen.

That last limit is the substantive one, and it is why the paper’s practical conclusion is not "use this index instead". It is that the family of diagnostics
1210 that can be run before deployment is the family that fails here, and the information that would order transfer is the information a practitioner does not have when the decision is made. The honest statement is that these twenty candidates were **not shown to order transfer**, not that they are shown incapable of it.

1215 5.5. *What the two interventions do and do not show*

Both interventions show the same shape. Pooling four regions never beats the best single-source transfer for any target and stays 0.28 to 0.50 below the within-region ceiling, so aggregation does not manufacture the missing conditional information. Removing the reversing predictors costs -0.081 of

1220 within-region skill with interval support in every region, and changes mean
transfer by +0.014 $[-0.017, +0.045]$, which is not distinguishable from zero.

That pair of numbers is easy to read as an exchange, and it is not one.
The thermal block's own contribution to transfer is +0.004, with an interval
spanning zero. Removing the reversing predictors returns +0.014, with an
1225 interval spanning zero. Both arms are null on the portability axis. What
the interventions measure is a local cost and no compensating transfer gain.
That is not a conservation law, and it is not a rate at which local skill can
be sold for portability. No such rate is estimated here.

5.6. The regime hypothesis, reported as it happened

1230 A regime-structure explanation was stated in advance and the data con-
firmed the null. The regime-distance correlation has the wrong sign at the
point estimate, the most regime-similar pair fails in both directions, and the
most regime-different pair transfers above chance. One mundane explanation
can be set aside: cutting Muğla, by far the largest population, to Manav-
1235 gat's cell count and separately to its positive count leaves Muğla's transfer
behaviour inside the subsampling range in both roles. The error was in the
hypothesised grouping, not in the data, and with ten pairs this cannot refute
regime typology [2] in general.

5.7. The empirical contrast with Dimarco et al.

1240 Dimarco et al. [8] transfer successfully across a comparable Mediterranean
design and we do not, and the two results are not in conflict. Their predictors
are attributes of a place, ours describe the state of a surface in one season,

which suggests that the relationship between domain similarity and transfer success is predictor-class dependent.

1245 That reading is one of at least two, and we cannot separate them here. Their response variable is human-driven **ignition**; ours is burned **area**. Ignition likelihood is dominated by human access and activity, stable and near-universal drivers, whereas burned extent is dominated by spread. A predictor-class explanation and a response-variable explanation are there-
1250 fore confounded in this comparison. Our own data speak against a simple predictor-class reading in any case: the static baseline transfers at a mean of 0.537 here, so within this cohort the place-attribute class does not travel either.

5.8. Implications

1255 For practice, one number should be stated before any other. In precision-recall terms, which is how a susceptibility surface is actually used, transferred models average a PR-AUC of 0.156 against a no-skill baseline of 0.136, and six of twenty directions fall below their own baseline. Whatever the ROC figures suggest, a model moved to a region it was not fitted in does not
1260 usefully rank burned cells there.

This paper also supports one concrete change in what is reported. Alongside a spatially blocked within-region figure, report skill on a held-out burn scar and its surroundings, or on a held-out fire event. On these five regions the two differ by about 0.14 ROC-AUC on the same model, which is the size
1265 of the effect such papers usually claim, so a blocked figure alone should be read as an upper bound. Transfer skill likewise has to be *measured* rather than inferred from similarity, and the pre-deployment diagnostics currently

used for that inference did not order it here. Where a model must be moved, the resource that closes the gap is target labels: thirty-two labelled 5 km
1270 blocks recover 85 to 89 % of the target’s matched ceiling in three of six directions, and 30 to 57 % in the rest. That is a real answer and not a cheap one, being 7 to 20 % of the target’s natural-vegetation population, and at small budgets the same intervention damages the direction that already transfers best.

1275 For method development, the results bound what unsupervised alignment can be asked to do. Two label-free methods applied carefully moved fourteen of twenty directions towards chance, and five of the six they moved upward involve Montiferru, the smallest region; the sixth, Manavgat to Muğla, moves downward. The committed-in-advance CORAL arm reaches a mean of 0.552,
1280 and taking whichever of the two methods scores better per direction reaches 0.556 — but that selection uses the target labels the protocol forbids, so it is an oracle upper bound rather than an achievable result (Section 4.3). Even the oracle only reaches the reference a model can reach on an unseen scar at all (Section 4.3), so they are not failing far below an achievable target; they
1285 are regressing the matrix onto it, which costs the directions that already worked. What alignment cannot do is exceed that reference, and a sign reversal is not a distribution mismatch that realigning inputs would repair.

5.9. Limitations

The transfer failure is a finding, not a limitation. The limitations are the
1290 boundaries on how far it generalises.

(i) **No meteorological covariates** enter the models, so we cannot say how local skill and portability behave for a mixed thermal-plus-weather pre-

dictor set.

(ii) **The same-geography comparison covers one region only**, and
1295 even there year and seasonal phase are confounded, a confound that cannot
be resolved in this study area for the reason given in Section 5.2. Its 331
burned cells also leave the thermal reversals unresolved at interval level,
and the pair holds place fixed but not population. Those two arms are
additionally the only transfer directions here computed by us rather than
1300 read from the pipeline author’s frozen export, with his unmodified code and
the same pinned environment.

(iii) **All labels derive from a single burned-area product**, MCD64A1
[10], whose omission and commission characteristics [4] bound every model
evaluated here. No second burned-area product covers 2021 and 2022 at this
1305 resolution; the companion paper reports what an independent active-fire ob-
servation says about the omission concern.

(iv) **Evia remains the most imbalance-atypical population** even
after the AOI extension, at a TSG prevalence of 0.287 against 0.038 to 0.072
in three of the others.

1310 (v) **Each region contributes one fire season**, so regional concept
shift is confounded with event meteorology, and distinguishing them requires
multi-year labels.

(vi) **Cross-region point estimates carry an implementation tol-
erance** of roughly ± 0.02 to 0.03 across scikit-learn versions. All reported
1315 numbers are fixed to one verified version, but exact reproduction elsewhere
requires the archived environment.

(vii) **The diagnostic correlations rest on an effective sample of**

ten region pairs. Both the successes and the failures of Section 4.4 should be read at that power.

1320 (viii) **Manavgat’s atypical transfer behaviour remains unexplained.**

It is where the conditional diagnosis bites hardest and where feature removal recovers most. Three candidates have now been tested and none survives: its meteorology, which was not extreme; the quality screening of its coarse thermal input, which propagates widely but moves no signed association by
1325 more than +0.0003 (Section 4.7, Appendix A(e)); and the evaluation frame, which explains its elevation figure but not its transfer behaviour (Section 4.9). With one fire season per region the remaining candidates are not separable in this design.

(ix) **The interval-support counts are less stable than the point**
1330 **estimates behind them.** Several verdicts sit within a thousandth of their reference value, and at 1 km blocking the published split of ten positive, seven negative and three uncertain turns on a lower bound of -0.00045 . The point estimates and the sign pattern are stable; the counts are not. Every sentence in this paper that leans on an exact count of supported directions
1335 should be read at that precision. (x) **The five areas of interest are not comparable frames, and this cohort cannot fully repair it.** Each region is a rectangle drawn around a fire, and the share of modelled cells lying beyond 10 km of any burned cell ranges from 2 % in Montiferru to 63 % in Bejís. Section 4.9 shows this is not cosmetic: equalising the frame to a 10
1340 km collar removes both supported elevation reversals, makes all five regions agree in sign on elevation, LST and TVDI, lifts mean transfer from 0.540 to 0.617 and reduces the below-chance directions from six to one. We report the

equalised arm alongside the frame-as-drawn arm rather than replacing one with the other, because the collar radius is itself a choice and 5 km and 10 km
1345 do not agree exactly (0.608 against 0.617). The deeper limitation is that the frames were fixed upstream of this work, in `repo/`, so we can restrict them but not extend them; a region whose rectangle is already fire-scale, Montiferru, cannot be given a far field for symmetry. Any future cohort should fix the frame by an explicit accessible-area rule [3] before any predictor is computed,
1350 and we treat that as the main design lesson of this paper.

(xi) **One classifier family.** The headline numbers use a random forest with unlimited depth, the configuration most able to encode local structure and least able to extrapolate. Appendix A(h) shows the transfer result is not an artefact of that choice: three further estimators, including a penalised
1355 linear one, all land between 0.510 and 0.556 and all place fourteen of twenty directions above chance. Those are point estimates without intervals, so the ordering among them is not claimed as a result. Other model families were not tried, and a different inductive bias might behave differently, but within this family the negative result is a property of the predictors rather than of
1360 an unregularised estimator.

6. Conclusions

Where a fire model is scored decides what it appears to know. Holding the model, its predictors and its fitting fixed and changing only which cells are scored, moving from a whole study region to the burn scar and its 2 km
1365 collar costs **0.143 ROC-AUC** [**+0.077**, **+0.208**]. A control isolates the cause: matching prevalence changes nothing, at -0.000 $[-0.003, +0.002]$,

while scoring the same predictions on the scar area costs +0.155 [+0.093, +0.217]. It is the composition of the negative pool, not class balance. The effect is larger than the predictor-block increments this literature publishes
1370 as findings, so a region-wide validation figure should be read as an upper bound on what a model achieves where fire actually occurs, and reported as one.

That correction is not only other people’s problem. Applied to our own five-region matrix it withdraws three of our claims. The study areas enclose
1375 very different far fields, from 2 % to 63 % of cells beyond 10 km of any fire, and that far field is higher ground. Equalising them to a 10 km collar, which drops no burned cells, lifts mean transfer from 0.540 to 0.617, reduces the directions below chance from six to one, and makes all five regions agree in sign on elevation, LST and TVDI. The sign reversal we had offered as the
1380 mechanism of the transfer residual is therefore mostly a property of how five rectangles were drawn. Under our own criterion no reversal then remains bootstrap-supported. The sign the five regions agree on is also not the one the dryness framing predicts: hotter pre-fire surfaces burned less everywhere, surviving stratification within elevation and greenness, so on this cohort the
1385 thermal block behaves as a proxy for fuel availability.

What survives the correction is the central result. Pre-fire thermal dryness adds a real and repeatable within-region increment, +0.06 to +0.15 ROC-AUC over a static and near-static baseline in all five regions, surviving spatial blocking at about 5 km and a predictor window closed up to
1390 two weeks before the first labelled burning. It does not travel. Equalised transfer of 0.617 against within-region skill near 0.87 leaves most of the gap

intact; the paired cross-region contribution is $+0.004$ $[-0.028, +0.036]$ with a sign that varies by pair; and the static baseline transfers at 0.537 against the thermal model’s 0.541, so this is not a peculiarity of dynamic predictors. Removing the two reversing predictors costs -0.081 of within-region skill, three quarters of it elevation, and changes transfer by $+0.014$, whose interval spans zero: a local cost is measured, no compensating transfer gain is, and no exchange between them is demonstrated.

The shortfall cannot be anticipated. None of the twenty diagnostics tested ordered the matrix on the marginal, niche or regime families, and on ten effective pairs those nulls mean not shown to order transfer rather than shown not to. The two that do carry intervals excluding zero measure signed direction agreement, rest on a data-selected subset of predictors, and require burned labels on both sides — which is the resource a practitioner deploying to a new region does not have. Label-free alignment compresses fourteen of twenty directions towards chance rather than repairing them, and pooled multi-region training does not escape it either. Supervised recalibration works but is not cheap: thirty-two labelled 5 km blocks recover 85 to 89 % of the target’s matched ceiling in three of six directions, and at the top budget the labelled set already holds most of one region’s burned cells.

Two practical consequences follow. Report local skill and portability separately, since a predictor block worth a large within-region increment may contribute nothing distinguishable from zero across regions. And fix the evaluation frame by an explicit accessible-area rule before any predictor is computed, because a frame chosen for convenience can manufacture both a transfer failure and a mechanism to explain it. Three extensions are left

for future work: temporal transfer, meteorological covariates, and physically normalised dryness variables. Each is a step towards the self-calibrating thermal monitoring system that motivated this study.

1420 **Appendix A. Appendix A. Sensitivity analyses**

Each arm below varies one design choice and leaves everything else fixed. None changes a conclusion in the main text. They are reported so that a reader can see which choices were tested, and what each was worth. Arm (f) is the exception in one respect: it is not a robustness check but a test of a
1425 claim the introduction makes, and the claim is not upheld.

(a) Evia AOI and prevalence. The raw transfer arms were repeated with the legacy, high-prevalence Evia box. Every qualitative conclusion is unchanged. Thermal raw transfer AUCs move by up to 0.07, the largest being Evia to Bejís at 0.378 to 0.448, and no direction changes side of the
1430 chance line.

(b) CORAL regularisation, including the canonical $\lambda = 1$. Over nine λ values from 0 to 10^{-1} on four directions, CORAL transfer AUC moves by at most 0.014 within any direction and 0.008 within the thermal family. The canonical $\lambda = 1$ of the cited method lies outside that sweep and was
1435 computed separately on the same four directions. It changes the thermal mean from 0.519 to 0.522, a shift of +0.003 with a per-direction range of -0.005 to $+0.016$, and none of those four directions crosses the chance line. The $\lambda = 1$ values are 0.510, 0.444, 0.559 and 0.575 against 0.508, 0.445, 0.564 and 0.559 at $\lambda = 0.1$.

1440 **Those four directions are Bejís–Muğla and Manavgat–Muğla,**

and they exclude the pair that moves. An earlier version of this appendix generalised from them to the whole matrix, which was an error: the Manavgat–Bejís pair is not in the sweep and it is the pair whose CORAL result is λ -dependent. A separate frozen artefact, `experiments/cross_`
1445 `region/step10/coral_lambda_sensitivity.csv`, carries it. Bejís to Manavgat gives 0.5571 [0.5292, 0.5857] at $\lambda = 10^{-5}$, 0.5644 at $\lambda = 10^{-3}$ and 0.5473 at $\lambda = 0.1$, all with intervals entirely above 0.5, but **0.4850 [0.4496, 0.5197] at $\lambda = 1$** , which crosses the chance line and loses its interval support. Manavgat to Bejís falls likewise, from 0.5108 to 0.4776 [0.4501, 0.5013].

1450 The correct statement is therefore narrower. CORAL-dependent conclusions are insensitive to λ over three orders of magnitude below 0.1; at the cited method’s canonical $\lambda = 1$ the one direction whose adapted interval sits entirely above chance loses that status. $\lambda = 10^{-5}$ is the minimally regularised choice and matches the reference implementation, but the dependence is real
1455 and is reported rather than absorbed.

(c) **Blocking scale.** Recomputing the transfer quantities at 10-cell ($\approx$ 5 km) blocking from the frozen per-cell predictions widens the intervals and moves the verdict counts, from ten positive, seven negative and three uncertain at 1 km to six, four and ten at 5 km. Coarser blocking therefore removes
1460 support from seven verdicts and adds none. Across five seeds the counts run 5 to 6 positive, 3 to 4 negative and 10 to 11 uncertain, so they are not exact.

The point estimates are unchanged. That is an identity rather than a result, and it should not be offered as robustness. The blocking scale is the bootstrap *resampling unit*, and each point estimate is computed once
1465 over all target cells, so no choice of block size could have moved one. The

comparison does establish two things. Every verdict that changes, changes towards "no verdict". And no direction crosses the chance line under the widened intervals. The counts are the fragile part of this paper. The sign pattern is not thereby shown to be robust. It is simply not tested by this variation.

(d) Predictor-window closure. The predictor window was closed 7 and 14 days earlier in all five regions. The thermal contribution stays positive and bootstrap-supported everywhere. The direction of the change is region-specific. It strengthens in Bejís, from 0.058 to 0.079 at 14 days, and in Muğla, from 0.115 to 0.128. It is flat in Montiferru. It weakens monotonically in Evia, from 0.156 to 0.149 to 0.135. What holds in every region is survival, not improvement. That is the claim made in Section 5.2.

(e) Quality screening of the coarse thermal input. Two of the five regions' MODIS inputs are quality-screened and three are not. The split follows export date rather than design, and it induces an elevation-correlated change at the input, at $r = +0.615$ in Manavgat. Manavgat's entire downstream chain was rebuilt from a quality-screened input. That changes the downscaled surface on 22,304 of 24,150 cells, by up to 10.9 °C. No signed univariate association moves by more than +0.0003. Elevation is identical in both arms, at 0.374, the population is unchanged, and the within-region increment moves from $[+0.055, +0.079]$ to $[+0.054, +0.077]$. The reason is structural. Elevation is a DEM variable the screening cannot touch, and fusion falls back on the MODIS-derived surface across only 2.14 percentage points of coverage. Details are in `paper/modis_qc_downstream_propagation.md`.

(f) **Normalised against absolute dryness channels.** Section 1.2 argues that an internally normalised index should be less exposed to absolute-temperature offsets between regions than raw land surface temperature. The thermal block contains both kinds, so the argument can be tested directly as
1495 a feature-set contrast.

One thing should be said before the result, because it makes the outcome less surprising than it might otherwise appear. The normalised channels are the two that hold their direction in the same-geography two-event comparison of Section 4.8, where the absolute channels move. That is a statement
1500 about one region across two fires. **Across regions it does not hold:** `1st_anomaly_mean` is itself one of the two bootstrap-supported reversing predictors, reversing between Bejís and Evia, which is why Section 4.6b drops it alongside elevation. Being internally normalised protects a channel against the offset between two seasons in one place. It does not, on this evidence,
1505 protect it against a change of place.

Three feature sets were run over all twenty directions and all five within-region folds, with the classifier, population, folds and bootstrap held fixed. The harness aborts unless its reference configuration lands on the frozen exports. It did exactly: the maximum absolute difference between the reference
1510 arm and the frozen transfer AUCs is 0.000000 across all twenty directions.

The advantage is not found. Per direction, the normalised set minus the absolute set has a mean of -0.0037 . It is positive in 12 of 20 directions and runs from -0.099 to $+0.094$, so the spread is an order of magnitude larger than the difference. The normalised channels transfer very slightly worse
1515 at the mean, and nothing here supports a design rule favouring anomaly-

Table A.15:

Feature set	Mean transfer AUC	Directions > 0.5	Mean within-region AUC
baseline only	0.5371	16	0.7896
baseline + normalised anomalies	0.5442	15	0.8551
baseline + absolute surface state	0.5479	13	0.8564
all ten features (reference)	0.5414	14	0.8883

referenced dryness for portability.

Within region the two sub-blocks are indistinguishable, at 0.8551 and 0.8564, and each recovers most of the gap between the baseline’s 0.7896 and the full model’s 0.8883. What separates them in sign stability does not become transferable skill.

This arm strengthens the negative finding rather than softening it. Four differently constituted feature sets were tried. None clears a mean of 0.548 across regions, and all differ sharply within them. The baseline arm also gives an independent confirmation of the control reported in Section 4.3: its mean transfer of 0.5371 was recomputed here from the modelling datasets and matches the 0.537 read from the frozen per-direction export. Details are in `paper/anomaly_only_transfer.md`.

The comparison is between two sub-blocks of one thermal set on one cohort. It does not test normalised dryness indices in general, and it does
1530 not test a normalisation fitted against a pooled multi-region reference rather than each region’s own baseline years.

(h) Model capacity. Every number in this paper comes from one random forest with unlimited depth and `min_samples_leaf = 3`. That is the configuration most able to encode local structure and least able to extrapolate, so a reader may reasonably ask whether the transfer failure is a property
1535 of the predictors or of the estimator. Holding the model fixed is right for the internal comparisons and does not license the external claim, so three further estimators were run over the same twenty directions and the same five within-region folds.

Regularisation does not rescue transfer. All four estimators land
1540 between 0.510 and 0.556, all four put exactly fourteen of twenty directions above chance, and the linear model, the one built to extrapolate, transfers worst. Within-region skill falls as capacity is reduced, from 0.888 to 0.741, which is what regularisation is expected to cost; transfer does not rise to
1545 meet it. The thermal block’s contribution to transfer is +0.004 under the canonical forest and **negative** under all three regularised alternatives, so the block does not become portable when the model is made simpler.

The transfer failure is therefore a property of the predictors on this cohort, not of an unregularised forest. Detail in `paper/model_capacity.json`.

(g) The coordinate-informed channels. `downscaled_lst_mean` and
1550 `fused_lst_mean` come from a per-region downscaling model whose own inputs include coordinates, which Section 3.14 names as the one route by which

Table A.16:

Estimator	Within-region AUC	Within increment	Transfer AUC	Transfer increment	Above chance
Random	0.888	+0.099	0.541	+0.004	14 of 20
for-					
est,					
depth					
un-					
lim-					
ited,					
leaf					
3					
Random	0.829	+0.055	0.556	−0.007	14 of 20
for-					
est,					
depth					
6,					
leaf					
50					
Random	0.805	+0.047	0.550	−0.021	14 of 20
for-					
est,					
leaf					
200					
Penalised	0.741	+0.045	0.510	−0.024	14 of 20
lo-					
gis-					
tic					
re-					
gres-					
sion					

a coordinate-derived surface re-enters a feature set that excludes coordinates. A coordinate-smoothed surface is by construction locally informative and
1555 non-portable, so if the within-region increment depended on it, that increment would be this paper’s own result in miniature rather than a finding about thermal dryness. The arm was therefore run here rather than deferred.

Both channels were dropped and everything refitted, over all twenty direc-
1560 tions and all five within-region folds. The reference configuration reproduces the frozen exports exactly, at 0.000000 against both the transfer and the within-region references.

Table A.17:

Region	Increment, full set	Increment without the two channels	Retained
Manavgat 2021	+0.067	+0.063	94 %
Bejís 2022	+0.056	+0.046	82 %
Muğla 2021	+0.116	+0.097	84 %
North Evia 2021	+0.153	+0.145	94 %
Montiferru 2021	+0.101	+0.105	103 %
Mean	+0.099	+0.091	92 %

The increment does not depend on them. It stays positive in every region and retains 82 % to 103 % of its size, rising slightly in one region. Mean
1565 transfer is likewise unchanged, at 0.545 without the two channels against 0.541 with them and 0.537 for the baseline alone. The local skill this paper reports is therefore not an artefact of a coordinate-smoothed surface.

This arm was computed independently here and reproduces the compan-

ion paper’s figure for the same quantity, 82 % to 103 %, from a separately
1570 written harness.

Appendix A.1. A(i). The four evaluations of Section 4.3, in full

Section 4.3 reports four evaluations of the same models as a ladder. The
per-split and per-scar detail is here, so that the ladder can be checked without
leaving the manuscript.

1575 **Table A1. Within-region half-split, every split.** Source and target
positive counts are given because they are unequal, which is the principal
limit on this arm: a straight cut does not produce two exchangeable halves.
Two splits are unusable because one half of Manavgat contains no burned
cells.

1580 **Table A2. Leave-one-scar-out at a 2 km buffer, every scar.** Muğla
and Montiferru are the only regions containing more than one burned compo-
nent of at least 50 cells. Muğla is the only region where holding one out still
leaves the source model properly trained across every arm; Montiferru’s two
components are very unequal, so one of its arms retains 472 source positives
1585 and the other 97. Muğla’s four arms mean 0.579; the four arms in the other
regions mean 0.525, and the pooled figure of 0.552 averages the two.

The evaluation populations of the two arms are not comparable with
each other or with the transfer targets. A held-out scar with its 2 km collar
contains only fire-adjacent negatives, against a whole target region’s inclusion
1590 of its easy far field; the burned fractions, 34 to 87 % against 3.8 to 28.7 %,
are a symptom of that rather than the cause, since ROC-AUC is invariant to
class balance at fixed class-conditional distributions. The scar arm therefore
asks for discrimination against the nearest and most similar negatives only,

Table A.18:

Region	Axis	Direction	Source positives	Target positives	Thermal AUC	Baseline AUC
manavgat 2021	east-west	low to high	700	84	0.695	0.661
manavgat 2021	east-west	high to low	84	700	0.557	0.580
manavgat 2021	north-south	low to high	0	784	—	—
manavgat 2021	north-south	high to low	784	0	—	—
bejis 2022	east-west	low to high	640	460	0.750	0.784
bejis 2022	east-west	high to low	460	640	0.578	0.560
bejis 2022	north-south	low to high	357	743	0.713	0.662
bejis 2022	north-south	high to low	743	357	0.603	0.613
mugla 2021	east-west	low to high	1,449	1,462	0.448	0.378
mugla 2021	east-west	high to low	1,462	1,449	0.636	0.628
mugla 2021	north-south	low to high	774	2,137	0.533	0.503
mugla 2021	north-south	high to low	2,137	774	0.294	0.549
evia 2021 extended	east-west	low to high	2,052	612	0.589	0.506
evia 2021 extended	east-west	high to low	612	2,052	0.645	0.482
evia 2021 extended	north-south	low to high	2,564	100	0.649	0.513
evia 2021 extended	north-south	high to low	100	2,564	0.533	0.495
montiferru 2021	east-west	low to high	410	129	0.475	0.461
montiferru 2021	east-west	high to low	129	410	0.507	0.513
montiferru 2021	north-south	low to high	438	101	0.583	0.511
montiferru 2021	north-south	high to low	101	438	0.536	0.447

Table A.19:

Region	Component	Source positives	Target positives	Target cells	AUC
		left			
evia 2021 extended	1	11	2,653	3,059	0.465
manavgat 2021	1	88	696	1,135	0.592
montiferru 2021	1	97	442	758	0.584
montiferru 2021	5	472	67	195	0.458
mugla 2021	6	1,997	914	1,266	0.595
mugla 2021	1	2,173	738	1,244	0.561
mugla 2021	10	2,272	639	940	0.617
mugla 2021	8	2,363	548	954	0.542

while a transfer arm includes the whole easy far field. Section 4.3 states the
 1595 consequence: the last three rows of the ladder are not distinguishable by this
 design.

Table A3. The foreign-region arm, decomposed by source. Each
 held-out scar area is scored with a model fitted on each of the other four
 regions. Table 5’s row D is the mean over the **eight** scars that carry a row
 1600 C, that is 32 of the 36 combinations below; the nine-scar mean quoted in this
 appendix is 0.559 against row D’s 0.555.

Table A.20:					
Target region	Scar	Mean over sources	Min	Max	Spread
Bejís 2022	1	0.589	0.521	0.631	0.110
North Evia 2021	1	0.590	0.374	0.704	0.329
Manavgat 2021	1	0.468	0.407	0.563	0.156
Montiferru 2021	1	0.504	0.457	0.546	0.089
Montiferru 2021	5	0.490	0.399	0.656	0.257
Muğla 2021	1	0.518	0.425	0.608	0.183
Muğla 2021	6	0.568	0.533	0.597	0.064
Muğla 2021	8	0.632	0.570	0.724	0.155
Muğla 2021	10	0.671	0.579	0.716	0.137

Over all 36 combinations here the mean is 0.559, the range 0.374 to 0.724,
 and ten fall below chance. Restricted to the 32 combinations behind row D
 the mean is 0.555 and the same ten fall below chance. The mean spread
 1605 across the four sources for a single scar is 0.164 over the nine scars and 0.171
 over the eight. Averaging over sources is what makes row D comparable with
 row C, which is fitted on one region; it is not a claim that the choice of

foreign source is immaterial, and Section 4.3 states the distinction.

Table A4. Prevalence is not the cause of the evaluation-area effect. The same fitted model and the same out-of-fold predictions are scored three ways: on the whole region, on a random sample of region cells drawn at the scar area’s own burned fraction, and on the scar area. Twenty draws per scar, seed 42. Nine scars, since this control needs no leave-one-scar-out arm and Bejís therefore qualifies.

Table A.21:

Target region	Scar	Burned fraction	A whole region	A' B scar area prevalence- matched
Manavgat 2021	1	0.61	0.797	0.796 0.589
Bejís 2022	1	0.66	0.824	0.825 0.575
Muğla 2021	6	0.72	0.777	0.772 0.564
Muğla 2021	1	0.59	0.777	0.779 0.654
Muğla 2021	10	0.68	0.777	0.779 0.673
Muğla 2021	8	0.57	0.777	0.782 0.761
North Evia 2021	1	0.87	0.864	0.866 0.745
Montiferru 2021	1	0.58	0.720	0.717 0.622
Montiferru 2021	5	0.34	0.720	0.723 0.461
Mean			0.782	0.782 0.627

A minus A', the effect of prevalence alone, is **−0.000** [−0.003, +0.002]. A' minus B, the effect of replacing the region’s negatives with fire-adjacent ones, is **+0.155** [+0.093, +0.217]. The evaluation-area effect is entirely the negative pool.

Appendix B. Appendix B. All twenty transferability diagnostics

1620 Section 4.4 summarises this table by family. The full ranking is given here because the negative result is the point: a reader should be able to see every candidate that was tried, not only the families' best members, and to check that no diagnostic was dropped after it failed.

Table B1. All transferability diagnostics versus raw thermal
1625 **transfer (20 ordered directions).** Spearman ρ with pair-based bootstrap 95 % CIs. Exp. = expected sign. Rows with $n = 12$ exist only for the four-AOI subset, because those diagnostics were never produced for Montiferru. The supported-features conditional rows use the 16 directions (8 pairs) with at least one CI-supported feature.

1630 *Table note.* A null row means the diagnostic was **not shown to order transfer** on this design, not that it was shown incapable of ordering it. With ten effective region pairs the power is low, and the intervals are wide enough to admit moderate true correlations in either direction. The last row is retained rather than deleted because it was computed: on two directions
1635 the statistic has no meaningful value, and reporting that is more honest than dropping the variant.

Reading the two rows that clear zero. Both are supported-feature variants, where the predictor subset is chosen by whether two regions' bootstrap intervals happen to be disjoint. That is a data-dependent selection
1640 made on the same data, uncorrected. Their all-nine-feature counterparts are in the table and both span zero. The result lives in the selection step, and Section 4.4 says so.

Equal-sample check. The families sit on unequal samples: marginal,

Table B.22:

Diagnostic	Family	Exp.	n dir	Spearman ρ [95% CI]	CI excludes 0
Agreement fraction, supported features	$P(y x)$ conditional	+	16	+0.84 [+0.58, +0.88]	yes
Cosine, supported features	$P(y x)$ conditional	+	16	+0.81 [+0.33, +0.88]	yes
Cosine, all 9 features	$P(y x)$ conditional	+	20	+0.50 [−0.17, +0.83]	no
Vector Spearman, all 9	$P(y x)$ conditional	+	20	+0.27 [−0.36, +0.77]	no
Agreement count, all 9	$P(y x)$ conditional	+	20	+0.18 [−0.40, +0.72]	no
Schoener’s D, 1-D mean	$P(x y=1)$ niche	+	20	+0.24 [−0.45, +0.74]	no
Warren’s I, 1-D mean	$P(x y=1)$ niche	+	20	+0.22 [−0.42, +0.73]	no
Schoener’s D, PCA-2D	$P(x y=1)$ niche	+	20	+0.10 [−0.51, +0.68]	no
Warren’s I, PCA-2D	$P(x y=1)$ niche	+	20	−0.07 [−0.66, +0.49]	no
Mahalanobis, burned centroids	$P(x y=1)$ niche	−	20	−0.23 [−0.75, +0.44]	no
Domain-classifier AUC	$P(x)$ marginal	−	20	−0.32 [−0.78, +0.33]	no
Predictor-space mean dissimilarity	$P(x)$ marginal	−	12	−0.10 [−0.54, +0.43]	no
Predictor-space p95 dissimilarity	$P(x)$ marginal	−	12	−0.08 [−0.59, +0.49]	no
Fraction inside weighted AoA	$P(x)$ marginal	+	12	+0.22 [−0.48, +0.59]	no
Fraction inside	$P(x)$ marginal	+	12	+0.08 [−0.89, +0.43]	no

applicability, climatic and geographic rows on twelve directions, the supported-
 1645 conditional rows on sixteen, the rest on twenty. Recomputing every row on
 the common twelve directions reproduces the published values to 4.8×10^{-5}
 and leaves the ordering unchanged — the conditional rows still lead at +0.87
 [+0.65, +0.88] and +0.85 [+0.43, +0.88], every marginal row still spans zero.
 Source: `paper/diagnostics_common_subset.md`.

1650 *Appendix B.1. B2. The signed univariate associations the reversal claim
 rests on*

Section 4.6b drops two predictors because their signed association with
 burning reverses between regions with bootstrap support on the frames as
 drawn. Section 4.9 shows that both supported elevation reversals disappear
 1655 once the frames are equalised, so this table is the evidence for the feature-
 removal arm and for the narrowed claim of Contribution 4, not for a general
 reversal mechanism. It was computed for the frozen analysis and is repro-
 duced here because the claim is otherwise asserted rather than shown.

Table B2. Signed univariate AUC of each predictor against
 1660 **burned, by region.** Primary natural-vegetation population; 10-cell (~ 5
 km) spatial-block bootstrap, 1000 replicates, seed 42. The AUC is never
 folded to $\max(\text{AUC}, 1 - \text{AUC})$, so a value below 0.5 means lower values
 rank burned and is a direction rather than weakness. **Bold** marks a region
 whose own interval excludes 0.5.

1665 **What counts as a reversal.** A pair of regions is called a reversal
 only when their signed associations point to opposite sides of 0.5 **and each**
region’s own interval excludes 0.5. That is stricter than requiring the
 two regions’ intervals to be disjoint, and the difference matters: for `current_`

Table B.23:

Feature	Manavgat	Bejís	Muğla	Evia	Montiferru
elevation_	0.374 [0.289,	0.643 [0.558,	0.611 [0.532,	0.541 [0.448,	0.584 [0.395,
mean	0.471]	0.729]	0.690]	0.626]	0.762]
slope_mean	0.531 [0.423,	0.521 [0.439,	0.637 [0.582,	0.487 [0.418,	0.652 [0.506,
	0.642]	0.605]	0.686]	0.554]	0.771]
ndvi_mean	0.636 [0.587,	0.559 [0.497,	0.662 [0.616,	0.639 [0.575,	0.586 [0.450,
	0.676]	0.619]	0.704]	0.701]	0.704]
lst_anomaly_	0.482 [0.428,	0.418 [0.364,	0.485 [0.395,	0.640 [0.567,	0.395 [0.285,
mean	0.530]	0.480]	0.566]	0.710]	0.535]
current_lst_	0.538 [0.452,	0.477 [0.401,	0.325 [0.271,	0.377 [0.301,	0.370 [0.248,
mean	0.621]	0.547]	0.382]	0.456]	0.513]
current_	0.552 [0.460,	0.517 [0.429,	0.336 [0.275,	0.362 [0.285,	0.356 [0.233,
tvdi_mean	0.641]	0.595]	0.398]	0.442]	0.499]
tvdi_	0.449 [0.391,	0.512 [0.443,	0.490 [0.396,	0.519 [0.444,	0.378 [0.282,
difference_	0.505]	0.583]	0.575]	0.589]	0.497]
mean					
downscaled_	0.552 [0.466,	0.484 [0.400,	0.307 [0.253,	0.377 [0.297,	0.365 [0.240,
lst_mean	0.637]	0.560]	0.366]	0.459]	0.511]
fused_lst_	0.540 [0.454,	0.481 [0.404,	0.325 [0.272,	0.376 [0.300,	0.370 [0.248,
mean	0.622]	0.551]	0.383]	0.456]	0.513]

1670 `1st_mean` between Manavgat and Muğla the two intervals are disjoint, but Manavgat’s own interval, [0.452, 0.621], includes 0.5, so Manavgat has no established direction to reverse from. That pair is a point reversal, not a supported one.

Table B3. The cross-region reversals that meet the stricter criterion. Difference intervals are from the same paired bootstrap.

Table B.24:					
Feature	Region A	AUC	Region B	AUC	Difference [95 % CI]
<code>elevation_mean</code>	Manavgat	0.374	Bejís	0.643	+0.269 [+0.138, +0.397]
<code>elevation_mean</code>	Manavgat	0.374	Muğla	0.611	+0.235 [+0.102, +0.360]
<code>1st_anomaly_mean</code>	Bejís	0.418	Evia	0.640	+0.221 [+0.123, +0.313]

1675 Three pair-level reversals across **two** features, which is why Section 3.12 removes exactly those two. Twenty-nine further pairs reverse at the point estimate only, spread across eight of the nine features, and they are not counted. The conservative criterion costs the paper findings rather than manufacturing them: a difference interval on the pair, which is the instrument
1680 Section 4.8 uses, would support more reversals than the three listed here.

Appendix B.2. B4. The transfer matrix in precision-recall terms

ROC-AUC is reported throughout the main text for comparability with the susceptibility literature. A susceptibility surface is used as a ranked

area budget, so precision-recall is the operational quantity, and at target
1685 prevalences of 3.8 to 28.7 % the two can differ sharply. Read from the same
frozen step9b exports as Table 7.

Table B4. Thermal transfer, PR-AUC against the no-skill base-
line. The baseline is the target’s burned prevalence. Lift is PR-AUC divided
by that baseline; a lift of 1 is no better than random ranking. Ordered by
1690 lift.

Six directions fall below their own no-skill baseline, and only one exceeds
twice it. The six are the same six that are below chance on ROC-AUC,
which is what a reversed ranking predicts in either metric. Section 4.9 shows
that this count is largely a property of the evaluation frames rather than of
1695 a reversed predictor-burning relationship.

Appendix C. Appendix C. Protocol detail

These protocols are given here in full rather than in Methods, because
each is a specification a reader needs only when checking the corresponding
result, and none is needed to follow the argument. Sections 3.2, 3.3, 3.11 and
1700 3.15 state in summary what each does.

Appendix C.1. C.1 Burned-area label, the reconstructed analysis grid, and the admissibility gate

Labels come from MODIS MCD64A1 Collection 6.1 (MODIS/061/MCD64A1)
retrieved through Google Earth Engine [11], whose omission and commission
1705 characteristics [4] bound every model here. The product is exported onto
the 30 m EPSG:4326 reference grid used by the rest of the pipeline, which

Table B.25:

Direction	ROC-AUC	PR-AUC	No-skill	Lift
Evia $\rightarrow$ Manavgat	0.686	0.094	0.038	2.45
Bejís $\rightarrow$ Montiferru	0.594	0.289	0.212	1.36
Evia $\rightarrow$ Montiferru	0.647	0.283	0.212	1.34
Bejís $\rightarrow$ Muğla	0.619	0.093	0.070	1.33
Muğla $\rightarrow$ Evia	0.653	0.379	0.287	1.32
Montiferru $\rightarrow$ Bejís	0.548	0.093	0.072	1.28
Montiferru $\rightarrow$ Muğla	0.619	0.089	0.070	1.27
Evia $\rightarrow$ Muğla	0.577	0.086	0.070	1.23
Muğla $\rightarrow$ Bejís	0.583	0.088	0.072	1.22
Manavgat $\rightarrow$ Evia	0.613	0.343	0.287	1.20
Manavgat $\rightarrow$ Montiferru	0.533	0.243	0.212	1.15
Montiferru $\rightarrow$ Manavgat	0.567	0.043	0.038	1.11
Montiferru $\rightarrow$ Evia	0.586	0.317	0.287	1.11
Muğla $\rightarrow$ Montiferru	0.531	0.214	0.212	1.01
Bejís $\rightarrow$ Manavgat	0.444	0.034	0.038	0.90
Manavgat $\rightarrow$ Muğla	0.470	0.063	0.070	0.90
Evia $\rightarrow$ Bejís	0.448	0.059	0.072	0.82
Bejís $\rightarrow$ Evia	0.383	0.226	0.287	0.79
Muğla $\rightarrow$ Manavgat	0.401	0.029	0.038	0.75
Manavgat $\rightarrow$ Bejís	0.326	0.049	0.072	0.67
Mean	0.541	0.156	0.136	1.16

duplicates each native ~ 500 m observation across the 30 m pixels beneath it.

The analysis grid is therefore **reconstructed rather than native**: the
1710 30 m grid is partitioned into non-overlapping blocks of `round(500 / 30) =`
17 $\times$ 17 pixels, giving a nominal cell edge of 510 m, and each cell is identified
by its integer block indices `row_500m` and `col_500m`. This approximates the
MODIS sinusoidal cell rather than reproducing it, being anchored to the
Landsat/EPSG:4326 reference grid, and marginal blocks are truncated. The
1715 cell is square in degrees but not on the ground, about 510 m north to south
and 390 to 407 m east to west, so its area is 0.199 to 0.208 km² against a
MODIS cell's 0.250 km²; every block-size label in this paper is therefore the
north-south dimension. The companion paper develops what follows from
that.

1720 A cell's representative burn date is the **mode** of the positive sub-pixel
day-of-year values in the block, tested against the region's label window. Be-
cause the exported raster carries no out-of-window positives, in this dataset
a single in-window positive sub-pixel makes the cell burned. The fraction of
positive sub-pixels agreeing with the modal date is recorded as `burn_date_`
1725 `pixel_agreement_fraction` but no agreement threshold is imposed. The
label never affects a cell's eligibility for modelling: unburned, all-no-data
and out-of-window cells all remain as the negative class.

Two safeguards are recorded rather than assumed. Cells that burned
inside a region's own predictor window are removed from its analysis universe
1730 rather than counted as unburned, which ran for Muğla, Evia and Montiferru,
is recorded as not run for Manavgat and has no recorded status for Bejís.

Burning in earlier years is screened for no region in the five-region cohort; the only historical exclusion in the study removes the 2021 Muğla scar from the 2022 event-relative experiment of Section 3.15.

1735 Before any modelling each region passes a gate that answers one question: of the cells labelled burned, what fraction is dominated by natural vegetation? A region is admitted as a wildfire candidate when that fraction reaches 0.50 and at least 30 burned cells are present, and is rejected as a cropland-dominated control when the cropland fraction reaches 0.50 instead.
1740 The gate uses ESA WorldCover classes aggregated to the same reconstructed cells. Its purpose is to separate burned area produced by natural-fuel combustion from burned area produced by post-harvest stubble burning, which MCD64A1 does not distinguish. Verdicts are reported in Section 4.1.

Appendix C.2. C.2 Transferability diagnostics versus transfer

1745 Twenty candidate diagnostics from five families are each rank-correlated against the same target quantity, the raw thermal transfer AUC over the twenty ordered directions, under one common pair-based bootstrap. The families are marginal predictor-distribution measures $P(x)$, burned-niche overlap $P(x|y=1)$, fire-regime label-pattern structure $P(y)$, and conditional feature-
1750 response direction $P(y|x)$.

The **marginal** family includes area-of-applicability-style dissimilarity in scaled, importance-weighted predictor space [23, 24, 18], climatic and geographic distance, and a learned domain classifier trained to separate source from target cells. The **niche-overlap** family includes Schoener’s D [36] and
1755 Warren’s I [44] over the burned cells of each region. The **regime** family includes burned-patch component counts and effective component counts. The

conditional family is the sign-agreement index: the fraction of predictors whose signed univariate association points the same way in both regions, computed over all predictors and over the subset whose associations are individually interval-supported.

Two properties of this design are stated in advance because they bound what it can show. The candidate set was fixed in a project analysis log before the correlations were computed; that log is not a formal pre-registration and no independent timestamped registration exists. And with five regions the effective sample is ten unordered pairs, so nineteen computed variants are evaluated at that power with no family-wise error control claimed; where a diagnostic clears an interval test but not a Bonferroni threshold, both are reported.

Appendix C.3. C.3 Same-geography event-to-event comparison (Muğla 2021 versus 2022)

Muğla is the one region for which a second fire event is analysed on the identical AOI and analysis grid, which allows the direction of feature-label associations to be compared with geography held fixed. The 2022 experiment is registered with windows anchored to its own event: a 58-day predictor window closing the day before ignition and a 49-day label window opening on it, matching the 2021 durations exactly.

This is not a clean temporal-transfer design and is not presented as one. The 2022 event ignites about six weeks earlier in the season than the 2021 event, so calendar year and seasonal phase are confounded and no difference can be attributed to the year alone. The registered claim is same-geography event-to-event. The pair is deliberately kept out of the twenty-

direction matrix, and that exclusion is enforced in released code rather than merely asserted here.

Two further properties bound the comparison. A historical-burn exclusion
1785 applies to the 2022 arm alone, masking every cell that burned in 2021 so the
previous year’s scar does not enter the following year’s analysis: it removes
3,073 cells, of which 2,941 belong to the primary population, taking it from
41,730 rows to 38,790, a drop of 7.0 %. The two arms therefore share 38,789 of
38,790 cells and three byte-identical static predictors, and in the 2022-to-2021
1790 direction every target positive lies outside the source training population.
Section 4.8 states what follows for how these two numbers may be read.

CRedit authorship contribution statement

Yunus Emre Cogurcu: Conceptualization, Methodology, Formal anal-
ysis, Investigation, Writing – original draft, Writing – review and editing,
1795 Supervision. **Emrehan Metin:** Software, Data curation, Investigation, Val-
idation, Writing – review and editing.

Declaration of competing interest

The authors declare that they have no known competing financial inter-
ests or personal relationships that could have appeared to influence the work
1800 reported in this paper.

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Data availability

1805 All satellite inputs are public and are obtained through Google Earth
Engine. The processing pipeline, its configuration and the frozen numeric
outputs are publicly available; the repository, the commit of record and the
licence are given in the data and code availability statement of the Methods,
together with the three components that fall outside that release. No digital
1810 object identifier is minted and no archival deposit exists.

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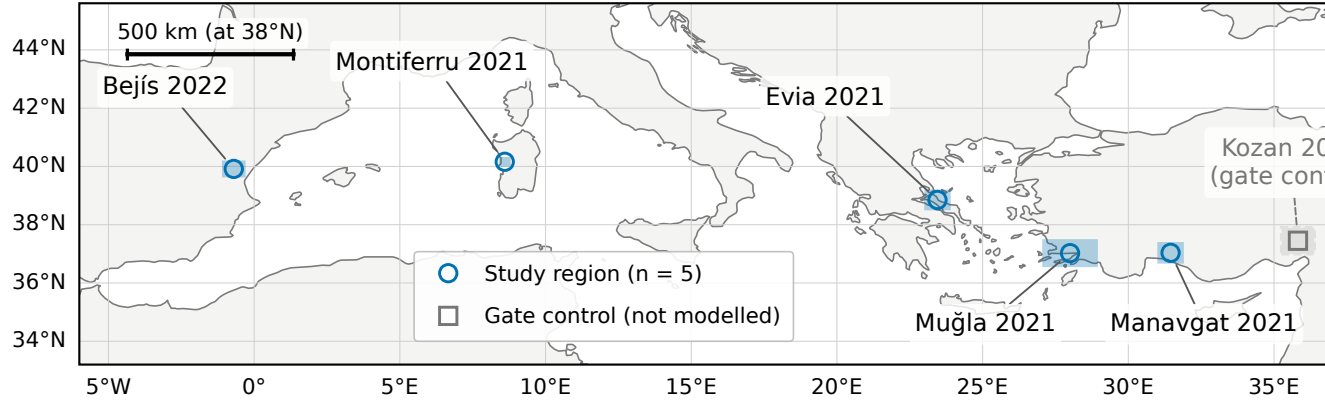


Figure C.1: **Study regions and the gate negative control.** Locations of the five wild-fire regions analysed in this study (blue, circular markers): Manavgat 2021 and Muğla 2021 (Türkiye), Bejís 2022 (Spain), North Evia 2021 (Greece, extended AOI) and Montiferru 2021 (Sardinia, Italy). Rectangles are the true analysis areas of interest; the ring marks each centroid, because at basin scale the rectangles alone are hard to locate. Kozan 2023 (grey, dashed outline, square marker) is *not* a study region: it is the negative control used to calibrate the burned-landcover admissibility gate, and it enters no model, transfer or diagnostic anywhere in this paper. Its rectangle is the true extent of its processed AOI and is larger than the study AOIs for that reason alone; the difference in size carries no methodological meaning and is not a design choice. Coastlines are Natural Earth 1:50 m; country borders are omitted deliberately. The scale bar is exact only at 38°N. Region bounding boxes match Table 1 to 5×10^{-3} degrees.

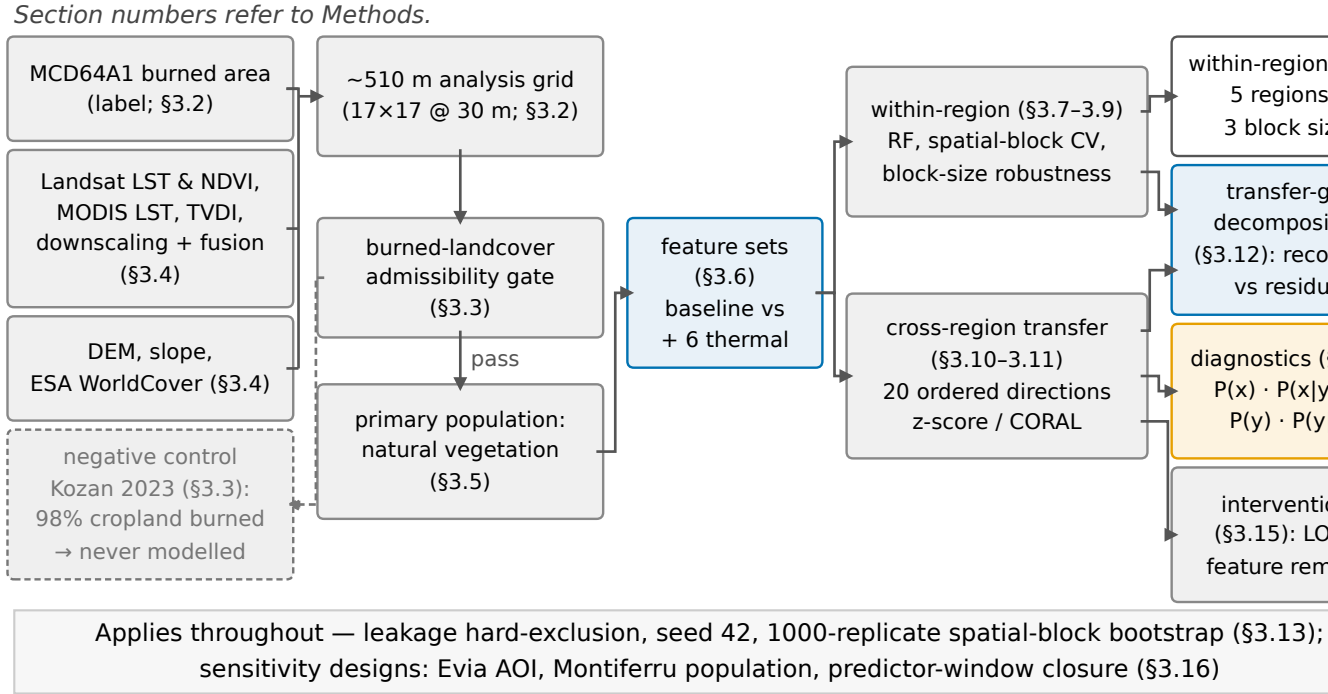


Figure C.2: Processing pipeline and evaluation programme. Flow from source products (left) to the transfer and diagnostic programme (right); every box carries the Methods subsection that specifies it. The five source products (MCD64A1 burned area, Landsat Collection 2 Level-2, MOD11A1, Copernicus DEM GLO-30 and ESA WorldCover) merge onto the reconstructed ~ 510 m analysis grid (17×17 pixels of the 30 m reference grid). The burned-landcover admissibility gate has two outcomes: regions that *pass* proceed to the natural-vegetation primary population, while the rejecting branch (grey, dashed) terminates in the Kozan 2023 negative control, whose burned area is 98% cropland and which is never modelled. The dashed connector deliberately carries no arrowhead onward, so the control cannot be read as entering the pipeline. Feature sets feed two evaluations in parallel: within-region (random forest under spatial-block cross-validation) and cross-region transfer (source-only, 20 ordered directions, with label-blind z-score and CORAL adaptation). Their levels are combined in the transfer-gap decomposition (§3.12). The footer states what applies throughout: leakage hard-exclusion, seed 42, 1000-replicate spatial-block bootstrap (§3.13), and the sensitivity designs (§3.16). Section references are verified against the Methods heading list when the figure is built.

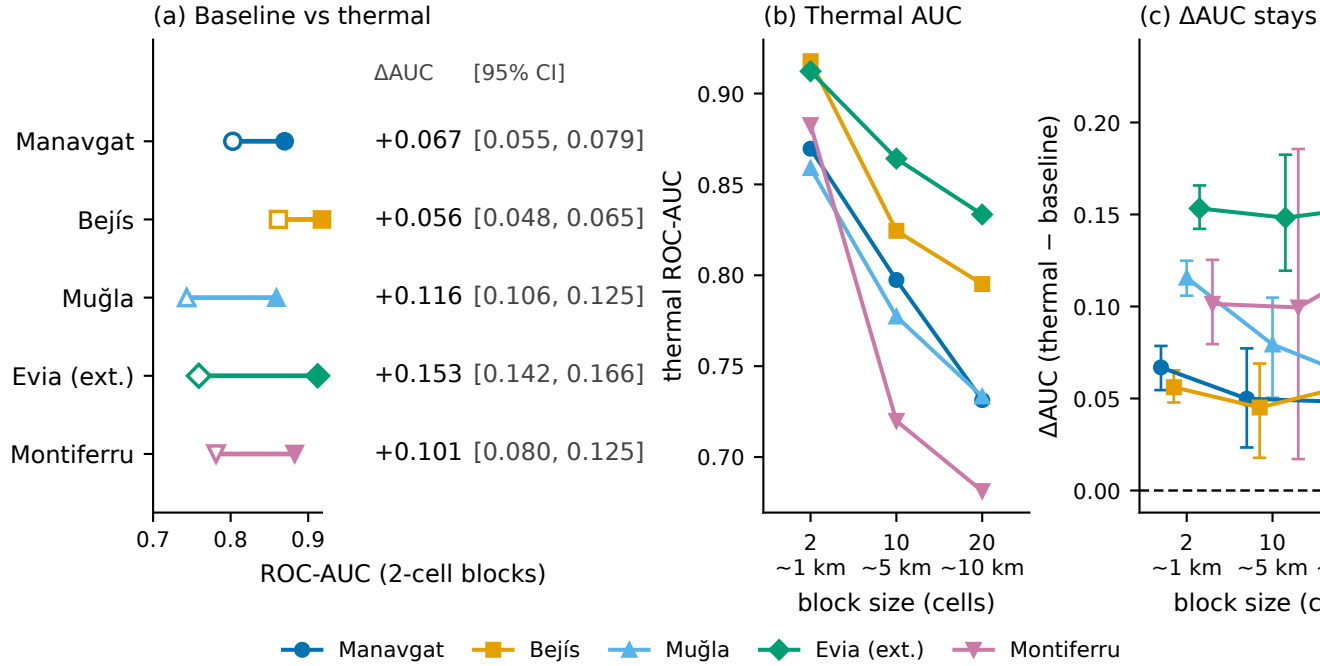


Figure C.3: **The thermal increment replicates in all five regions and survives coarser spatial blocking.** Natural-vegetation primary population; random forest with the primary weighted configuration; all intervals are 95% spatial-block bootstrap intervals (1000 replicates, seed 42). **(a)** Baseline versus thermal out-of-fold ROC-AUC per region at the default 2-cell (~ 1 km) blocking: the open marker is the static baseline feature set, the filled marker is the baseline plus the six thermal predictors, and the connecting bar is the increment. The right-hand column reports the paired ΔAUC and its 95% interval; it is a fixed column placed beyond the plotted AUC range, so no label position depends on a data value. Every interval excludes zero. **(b)** Absolute thermal ROC-AUC against spatial block size (2, 10 and 20 cells, $\approx 1, 5$ and 10 km): absolute skill falls as blocking coarsens, in every region. **(c)** The thermal increment ΔAUC against the same block sizes, with 95% intervals: the increment does *not* melt away, and its interval excludes zero in all ten of the 2- and 10-cell region-block-size combinations. The five 20-cell intervals also lie above zero but rest on 6 to 33 positive-carrying blocks each, so they are plotted as indicative rather than as intervals (Table 3, table note). In (c) the five regions are offset horizontally by 0.15 cell so their whiskers separate; **this offset is cosmetic, since all five regions are evaluated at exactly the same three block sizes** (2, 10, 20 cells), and the offset is smaller than half the spacing between block sizes, so the groups cannot merge. Colour and marker shape both encode region, so the panels remain readable in greyscale.

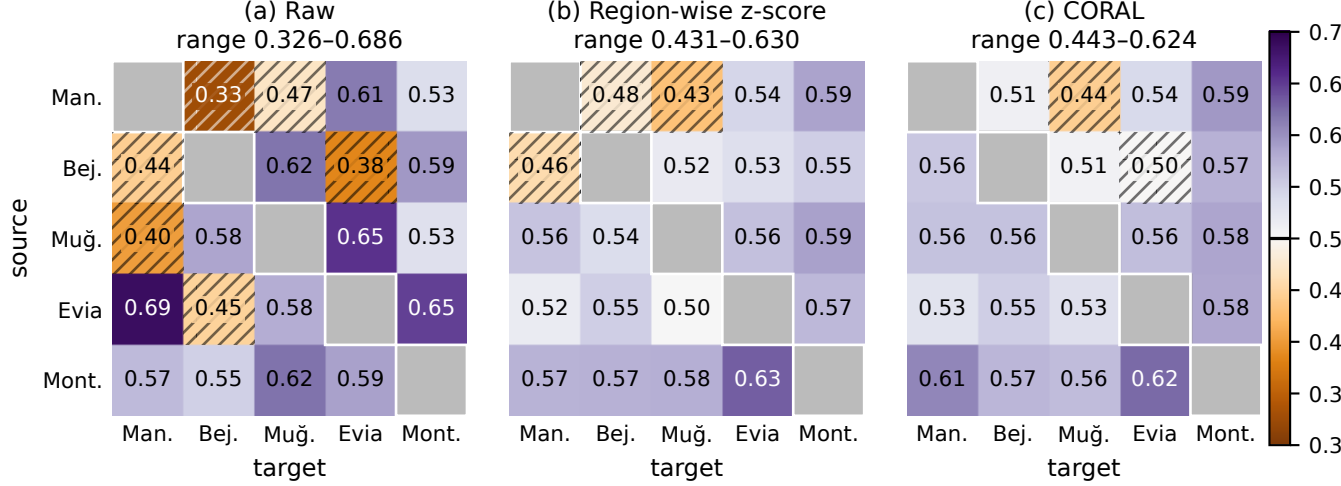


Figure C.4: **Cross-region transfer, and what label-blind adaptation does to it.** Thermal transfer ROC-AUC for all 20 ordered source–target directions among the five regions, natural-vegetation primary population. (a) Raw transfer, no adaptation. (b) After region-wise z-score standardisation. (c) After CORAL. Rows are the source region, columns the target; the grey diagonal is within-region performance and is *not* a transfer result. The diverging colour scale is centred exactly on chance (0.5, marked by the black rule on the colour bar), so orange is below chance and purple above. Each panel header states that panel’s range: adaptation compresses the spread from 0.326–0.686 (raw) to 0.431 to 0.630 (z-score) and 0.443 to 0.624 (CORAL). It pulls the extremes toward chance rather than lifting transfer. Cells below chance are hatched. The hatch is a redundant cue for print: a diverging palette is symmetric in luminance about its centre, so in greyscale a below-chance cell would otherwise be indistinguishable from an equally distant above-chance one. Hatching follows the *exact* value while the printed label is rounded to two decimals, so two cells print “0.50” with different hatching: CORAL Bejís→Evia is 0.4991 (hatched, below chance) and z-score Evia→Muğla is 0.5013 (un-hatched, above chance). Every cell carries its value, and every label’s colour is chosen by measured WCAG contrast against that cell’s rendered colour (worst case 4.7:1). Confidence intervals for these directions are given in Table 4.

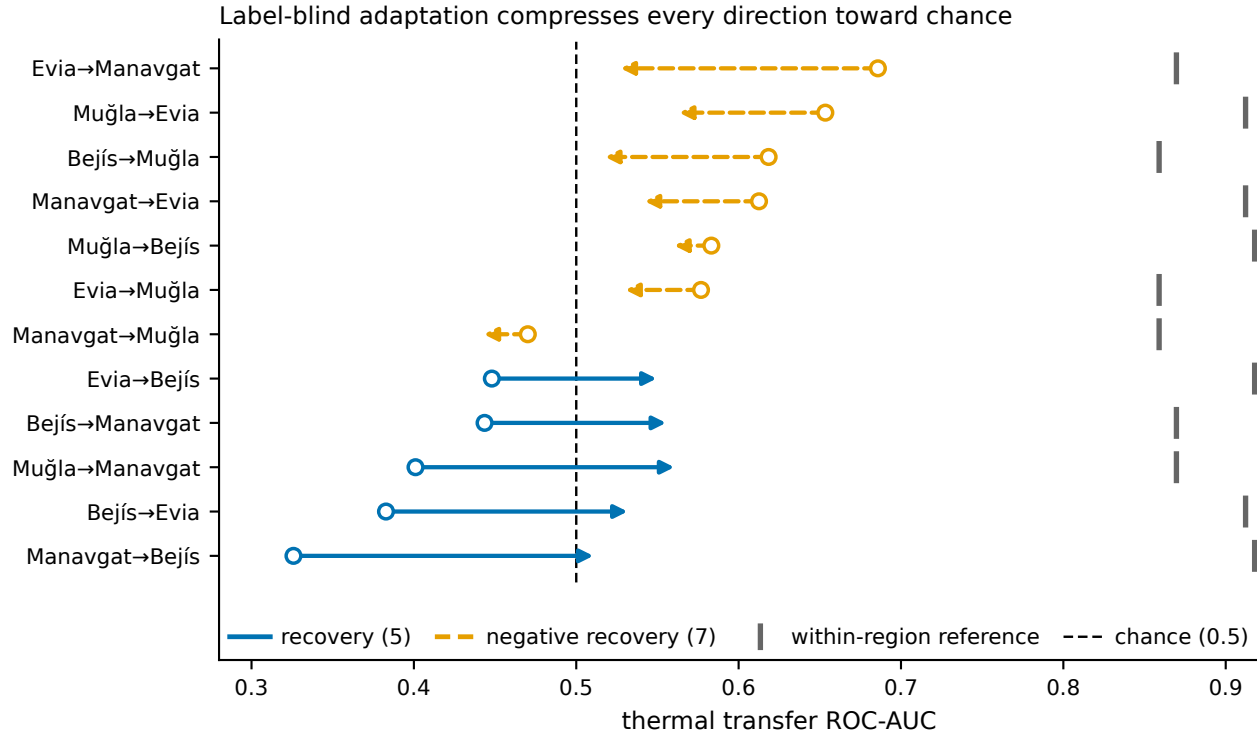


Figure C.5: **Label-blind adaptation regresses transfer onto the level a model reaches on an unseen scar, rather than recovering it.** All 12 ordered directions among the four regions carrying the full decomposition (Manavgat, Bejís, Muğla, Evia), natural-vegetation primary population. Each row is one direction. The open circle is raw transfer ROC-AUC and the arrow runs to the best label-blind adapted result for that direction (region-wise z-score or CORAL, whichever is better); the grey vertical tick is the within-region reference for that target, and the dashed vertical rule is chance. **Solid blue arrows are the five directions where adaptation moves transfer toward the within-region reference (recovery); dashed orange arrows are the seven where it moves *away* from it (negative recovery).** Line style and colour both encode this class, because the two hues differ by only 2.30:1 in relative luminance and would nearly merge in greyscale. The pattern is the point: all six directions starting above chance are pushed down, and five of the six starting below chance are pushed up, so adaptation acts on the spread rather than on skill. The exception is Manavgat→Muğla, which starts below chance at 0.470 and is pushed further down to 0.443. The worst case is Evia→Manavgat, where raw transfer of 0.686 is dragged down to 0.527 against a within-region reference of 0.870, a recovered fraction of -0.862 . No arrow reaches its within-region tick.

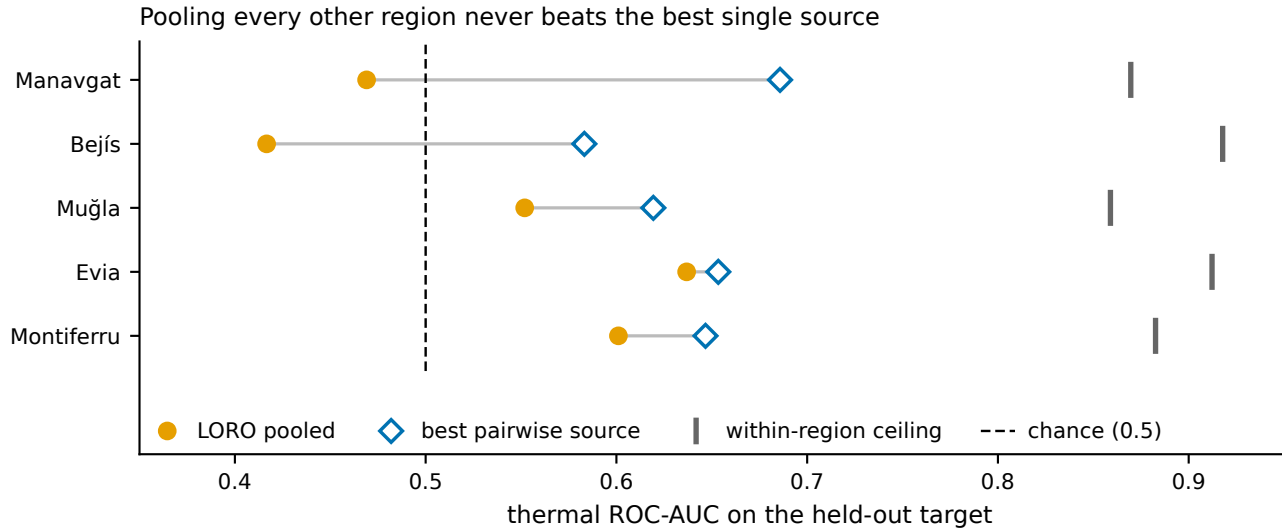


Figure C.6: **Pooling every other region never beats the best single source.** Leave-one-region-out evaluation: for each held-out target region (rows), a model trained on the pooled remaining four regions (filled orange circle) is compared against the best individual source region for that same target (open blue diamond), with the within-region ceiling as a grey vertical tick and chance as the dashed rule. Natural-vegetation primary population, thermal feature set, no adaptation. The pooled model lies to the left of the best pairwise source in all five targets without exception. Adding source regions does not accumulate transferable signal. In two targets the pooled model falls *below chance* at the point estimate (Manavgat 0.469, Bejís 0.417), and only for Bejís does the interval lie entirely below chance, while a single well-chosen source reaches 0.686 and 0.583 respectively. Every point remains far below the within-region ceiling.

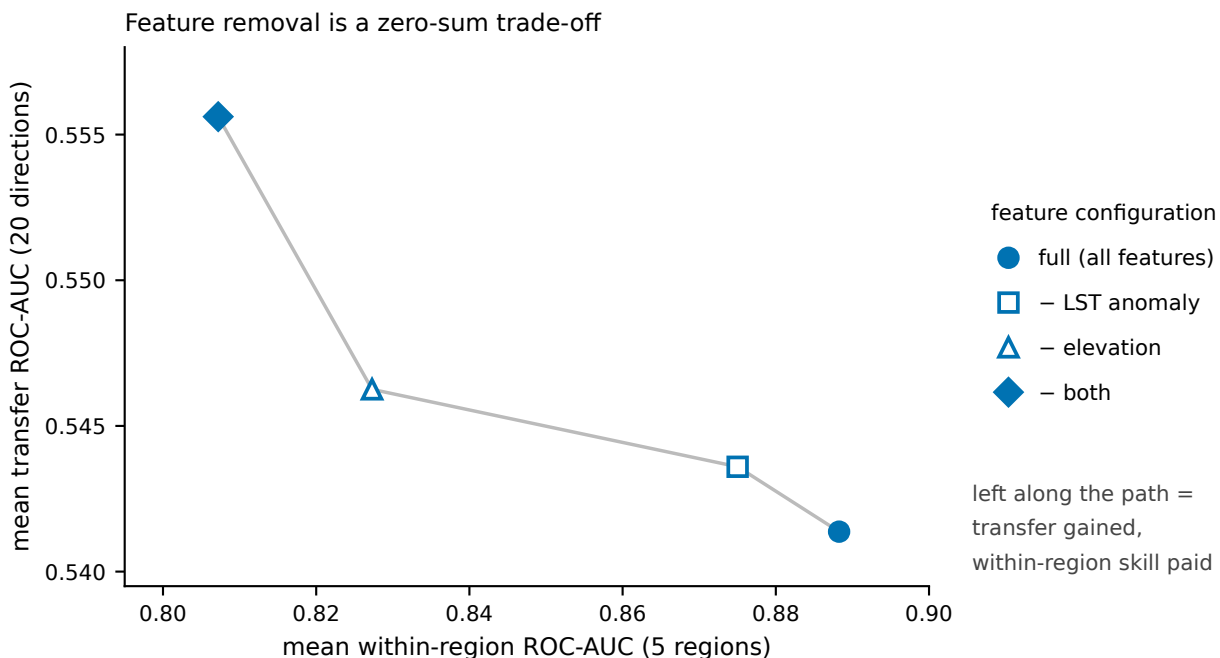


Figure C.7: **Removing the direction-reversing features costs locally and returns nothing measurable on transfer.** Mean within-region ROC-AUC across the five regions (horizontal) against mean transfer ROC-AUC across the 20 ordered directions (vertical), for four feature configurations: the full thermal set, the set without the LST anomaly, without elevation, and without both. Each configuration has its own marker and is named in the key; the grey path connects them in that order. Moving left along the path, transfer rises monotonically ($0.541 \rightarrow 0.556$) while within-region skill falls monotonically ($0.888 \rightarrow 0.807$). Both monotonicities are verified when the figure is built, so the figure cannot silently contradict the claim. The within-region cost is 0.081 AUC and is supported in every region. The change in transfer is +0.014 AUC on an interval that spans zero, so no exchange between the two is demonstrated: a local cost is measured and no compensating transfer gain is.

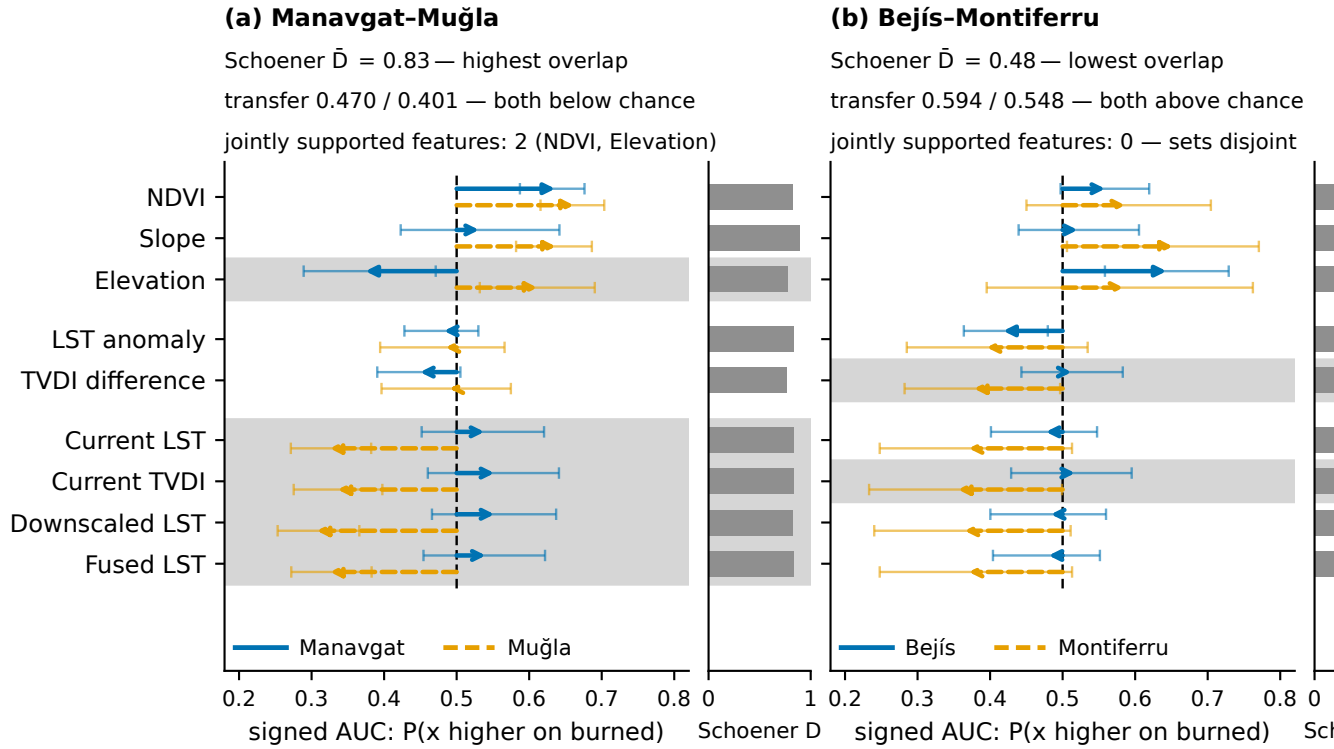


Figure C.8: **Burned-niche overlap does not determine transfer: the contrast pair.** Per-feature signed univariate AUC, the probability that a predictor takes a higher value on a burned cell than on an unburned one, for the two most informative region pairs, natural-vegetation primary population. Arrows run from chance (0.5) to the observed value, so their direction is the sign of the association and their length its strength; whiskers are 95% spatial-block (≈ 5 km) bootstrap intervals. Within each pair the two regions are distinguished by colour, by line style (solid for the first region named, dashed for the second) and by vertical offset; line style is included because the two hues differ by only 2.30:1 in relative luminance and would nearly merge in greyscale. **A filled arrowhead means that region's 95% interval excludes 0.5, so the direction is statistically supported; an open arrowhead means the interval contains 0.5 and the direction is not supported.** Rows shaded grey are features whose sign disagrees between the two regions. Grey side bars are the per-feature Schoener's D overlap. **(a)** Manavgat–Muğla, the pair with the *highest* burned-niche overlap ($\bar{D} = 0.83$), transfers below chance in both directions (0.470 and 0.401). **(b)** Bejís–Montiferru, the pair with the *lowest* overlap ($\bar{D} = 0.48$), transfers above chance in both directions (0.594 and 0.548). High niche overlap is therefore not sufficient for transfer. The arrowhead fill also exposes the coverage limit of the supported-agreement index, which is defined only on features whose direction is supported in *both* regions: there are two such features in (a) (NDVI and elevation) and **none in (b)**, where the two regions' supported sets are disjoint and Montiferru's intervals

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